COUNTRY PROFILES

Nuclear Power in China

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- The impetus for nuclear power in China is due to air pollution from coal-fired plants, as well as climate commitments and energy security.
- China's policy is to have a closed nuclear fuel cycle.
- China has become largely self-sufficient in reactor design and construction, but is making full use of Western technology while adapting it.
- Relative to the rest of the world, a major strength is the nuclear supply chain.
- China's policy is to 'go global' with exporting nuclear technology including heavy components in the supply chain.

61

OPERABLE REACTORS

60,289 MWe

38

REACTORS UNDER CONSTRUCTION

39,958 MWe

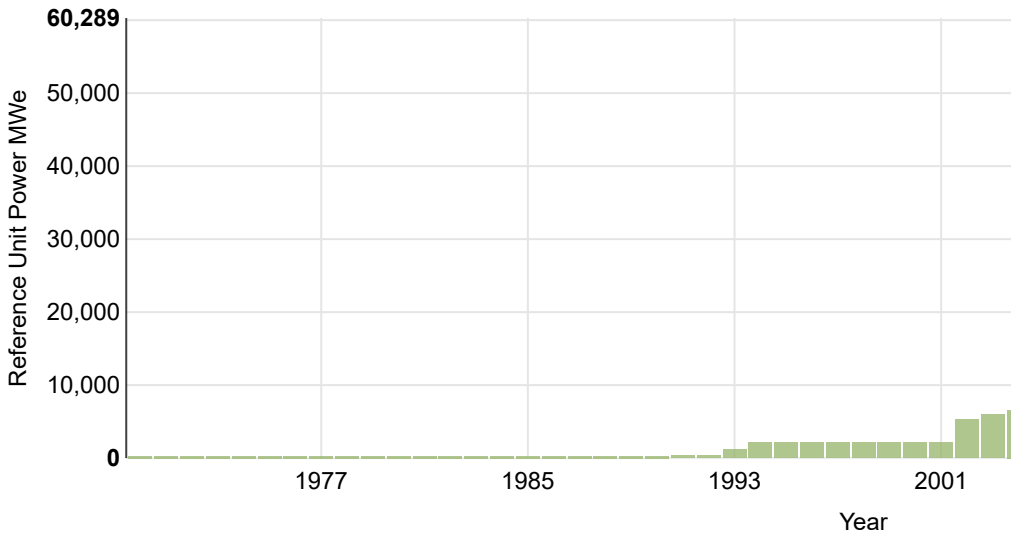
0

REACTORS SHUTDOWN

0 MWe

Privacy - Terms

Operable nuclear power capacity



Electricity sector

Total generation (in 2023): 9548 TWh

Generation mix: coal 5857 TWh (61%); hydro 1286 TWh (13%); wind 886 TWh (9%); solar 584 TWh (6%); nuclear 435 TWh (5%); natural gas 283 TWh (3%); biofuels & waste 206 TWh (2%).

Import/export balance: 17.1 TWh net export (5.0 TWh imports; 22.1 TWh exports)

Total consumption: 8013 TWh

Per capita consumption: c. 5700 kWh in 2023

Source: International Energy Agency and The World Bank. Data for year 2023.

Rapid growth in demand has given rise to power shortages, and the reliance on fossil fuels has led to much air pollution. The economic loss due to pollution is put by the World Bank at almost 6% of GDP,[1] and the new leadership from March 2013 prioritized this.* Chronic and widespread smog in the east of the country is attributed to coal burning.

* Official measurements of fine particles in the air measuring less than 2.5 micrometres, which pose the greatest health risk, rose to a record 993 micrograms per cubic metre in Beijing on 12 January 2013, compared with World Health Organization guidelines of no higher than 25.

In August 2013 the State Council said that China should reduce its carbon emissions by 40-45% by 2020 from 2005 levels, and would aim to boost renewable energy to 15% of its total primary energy consumption by 2020. In 2012 China was the world's largest source of carbon emissions – 2626 MtC (9.64 Gt CO_2), and its increment that year comprised about 70% of the world total increase. In March 2014 the Premier said that the government was declaring "war on pollution" and would accelerate closing coal-fired power stations.

In November 2014 the Premier announced that China intended about 20% of its primary energy consumption to be from non-fossil fuels by 2030, at which time it intended its peak of CO_2 emissions to occur. This 20% target is part of the 13th Five-Year Plan and was reiterated at the Paris climate change conference in December 2015, along with reducing CO_2 emissions by 60-65% from 2005 levels by 2030. In September 2020, the Premier stated that the country now intends to reach peak emissions before 2030, and to become carbon neutral before 2060.

In the 13th Five-Year Plan (2016-2020) for power production announced by the National Energy Administration (NEA) in November 2016, coal capacity was to be limited to 1100 GWe by 2020 by cancelling and postponing about 150 GWe of projects. Gas in 2020 was projected at 110 GWe, hydro 340 GWe, wind 210 GWe, and solar 110 GWe, of which distributed PV was to be 60 GWe. Nuclear 58 GWe was reiterated for 2020.

China's 14th Five-Year Plan (2021-2025) set a target to reach 70 GWe gross of nuclear capacity by the end of 2025. By the end of 2025, operating capacity reached about 62 GWe, with a further 43 GWe under construction.

China's draft 15th Five-Year Plan (2026-2030), published in March 2026, targets 110 GWe of nuclear capacity by 2030.

The grid system run by the State Grid Corporation of China (SGCC) and China Southern Power Grid Co (CSG) is sophisticated and rapidly growing, utilizing ultra high voltage (1000 kV AC and 800 kV DC) transmission. In 2024, SGCC invested about CNY 600 billion on the grid system.

The main nuclear operators are China National Nuclear Corporation (CNNC) and China General Nuclear Power Group (CGN), with SPIC (via its nuclear power business State Nuclear Power Technology Corporation – SNPTC) a third one.

Energy policy and clean air

While coal is the main energy source, most reserves are in the north or northwest and present an enormous logistical problem – nearly half the country's rail capacity is used in transporting coal. Because of the heavy reliance on old coal-fired plant, electricity generation accounts for much of the country's air pollution, which is a strong reason to increase nuclear share. China is by far the world's largest contributor of carbon dioxide emissions. Gas consumption in 2025 is expected to exceed 450 billion cubic metres according to China National Petroleum Corp. Of that, just over half will be produced domestically. The Power of Siberia pipeline connects Russia to China's south and east and has export capacity of 38 billion cubic metres per year.

Water is also a constraint on coal-fired power generation, much of which is in water-deficient regions. Retrofit to air cooling decreases efficiency by 3-10% and is reported to cost about $200 million per GWe in China.

Desalination is an important part of China's long-term water strategy. The country's lack of fresh water, and overuse of underground water resources, poses significant health problems for a minority of its population. China produces about 10 million m^3/day of desalinated water at present, about 10% of the world's total, and is building further capacity.

In September 2014 a national climate change plan prepared by the NDRC was approved by the State Council. This set emissions and clean energy targets for 2020. The carbon emissions intensity target was set at 40-45% reduction from 2005 to 2020. It aims to increase the shares of non-fossil fuels in primary energy consumption to about 15% by 2020 – at the end of 2018 it was 12%. The plan also set the target for China to increase forest coverage by 40 million hectares by the end of 2019. The government said it would speed up efforts to establish a carbon emissions permit market, as well as deepening international cooperation under the principles of "common but differentiated responsibilities", equity, and respective capability.

The State Council published the *Energy Development Strategy Action Plan, 2014-2020* in November 2014. The plan aimed to cut China's reliance on coal and promote the use of clean energy, confirming the 2012 target of 58 GWe nuclear in 2020, with 30 GWe more under construction. The plan called for the "timely launch" of new nuclear power projects on the east coast and for feasibility studies for the construction of inland plants. It said that efforts should be focused on promoting the use of large pressurized water reactors (including the AP1000 and CAP1400 designs), high temperature gas-cooled reactors (HTRs) and fast reactors. It also said that research should be conducted to "improve the nuclear fuel cycle system" including reprocessing of used fuel.

In the **13th Five-Year Plan** from 2016, six to eight nuclear reactors were to be approved each year. Non-fossil primary energy provision should reach 15% by 2020 and 20% by 2030 (from 9.8% in 2013). The action plan aim was for 62% of primary energy to come from coal in 2020, down from 72.5% in 2007. This was achieved in 2018, though largely through substitution for natural gas.

In June 2015 China submitted its Intended Nationally Determined Contribution (INDC) to climate change mitigation and adaptation for 2020 to 2030 to the UN. This pledge included increasing the share of non-fossil fuels in primary energy consumption to about 20% and restraining carbon emissions in 2030 to double of those in 2005 (after being 158% of the 2005 level in 2015 and 182% in 2020). Annual average new nuclear capacity 2005 to 2020 is 3.4 GWe/yr, from 2020 to 2030 it is 9.0 GWe/yr.

In November 2025 China submitted its 2035 Nationally Determined Contribution to the United Nations Framework Convention on Climate Change (UNFCCC), committing to reduce economy-wide net greenhouse gas emissions by 7-10% from peak by 2035 and to raise the non-fossil energy share to over 30% of total energy.

Nuclear power

Nuclear power has an important role, especially in the coastal areas remote from the coalfields and where the economy is developing rapidly. Generally, nuclear plants can be built close to centres of demand, whereas suitable wind sites are remote from demand. Moves to build nuclear power commenced in 1970 and about 2005 the industry moved into a rapid development phase, in the 11th Five-Year Plan.

Technology has been drawn from France, Canada and Russia, with local development based largely on the French element. The latest technology acquisition has been from the USA (via Westinghouse, owned by Japan's Toshiba) and France. The State Nuclear Power Technology Corporation (SNPTC) made the Westinghouse AP1000 the main basis of technology development in the immediate future, particularly evident in the local development of CAP1400 based on it, and more immediately the CAP1000.

This has led to a determined policy of exporting nuclear technology, based on China's development of the CAP1400 reactor with Chinese intellectual property rights and backed by full fuel cycle capability. In 2015 the Hualong One reactor became the main export product. The policy is being pursued at a high level politically, as one of 16 key national science & technology projects, utilising China's economic and diplomatic influence, and led by the initiative of CGN commercially, with SNPTC and more recently CNNC in support.

Workers bringing Ling Ao Phase II unit 1 to first criticality in July 2010 (CGN)

Prior to 2008, the government had planned to increase nuclear generating capacity to 40 GWe by 2020 (out of a total 1000 GWe planned), with a further 18 GWe nuclear being under construction then. Projections for nuclear power then increased to 70-80 GWe by 2020, 200 GWe by 2030 and 400-500 GWe by 2050. In April 2015 CNEA said that by 2030 per capita annual electricity consumption would be 5500 kWh, and installed nuclear capacity would be 160 GWe, providing 10% of electricity (with coal 64.6%). By 2050, per capita consumption would be 8500 kWh, and installed nuclear capacity 240 GWe providing 15% of electricity (coal 50.5%).

National policy moved from 'moderate development' of nuclear power to 'positive development' in 2004, and in 2011-12 to 'steady development with safety'. See further comment under Post-Fukushima Review below. The nuclear capacity target for 2020 became 58 GWe in operation and 30 GWe under construction, though the China Electricity Council in 2019 said that 53 GWe in operation was more likely.

In December 2011 the National Energy Administration (NEA) said that China would make nuclear energy the foundation of its power-generation system in the next "10 to 20 years", adding as much as 300 GWe of nuclear capacity over that period. In September 2013 SNPTC estimated that 4-6 new units per year would be needed to 2015 then 6-8 units during the 13th Five-Year Plan period (2016-2020)*, increasing to 10 units each year after 2020. The NEA confirmed that China could manufacture eight full sets of reactor equipment per year, and in 2014 it announced that China was aiming for world leadership in nuclear technology.

* The **13th Five-Year Plan** formalized in March 2016 included the following nuclear projects and aims:

- Complete four AP1000 units at Sanmen and Haiyang.
- Build demonstration Hualong One reactors at Fuqing and Fangchenggang.
- Start building the demonstration CAP1400 reactor at Rongcheng (Shidaowan).
- Accelerate building Tianwan Phase III (units 5&6).
- Start building a new coastal power plant.
- Active preparatory work for inland nuclear power plants.
- Reach target of 58 GWe nuclear operational by end of 2020, plus 30 GWe under construction then.
- Accelerate and push for building demonstration and large commercial reprocessing plants.
- Strengthen the fuel security system.

In July 2013 the National Development and Reform Commission (NDRC) set a wholesale power price of CNY 0.43 per kWh (7 US cents/kWh) for all new nuclear power projects, to promote the

healthy development of nuclear power and guide investment into the sector. The price is to be kept relatively stable but will be adjusted with technology advances and market factors, though many consider it not high enough to be profitable. It was reported that the price for power from Sanmen might be about 5% higher, but in 2019 it was CNY 0.42/kWh. Haiyang was selling power for CNY 0.414/kWh and Taishan CNY 0.435/kWh in 2019.

Nuclear power is already competitive, and wholesale price to grid has been less than power from coal plants with flue gas desulfurization, though the basic coal-fired cost is put at CNY 0.3/kWh*. In March 2015 a new round of electricity market reform was launched, to prioritize clean power generation, and this allowed nuclear power companies to negotiate prices with customers.

* Wind cost to grid is CNY 0.49 - 0.61 per kWh, depending on region (and FIT, now CNY 0.54 per kWh), solar is CNY 0.9 (desert) to 1.3 (east). CGN Power reported that in 2015 Hongyanhe grid tariff was CNY 0.4142/kWh, and Ningde CNY 0.43/kWh.

In October 2018 the NDRC's Energy Research Institute said that China's nuclear generating capacity must increase to 554 GWe by 2050 if the country is to play its part in limiting the global temperature rise to below 1.5 °C. The share of nuclear power in the country's energy mix would thus increase from 4% to 28% over this period. The study said that assuming an all-in cost of CNY 20,000 (approximately $3000) per kW of capacity in large plants, an investment of more than CNY 8.7 trillion ($1.3 trillion) would be required. Based on capacity additions over the past few years, the total investment demand to 2050 was considered to be feasible.

Hong Kong supply

Hong Kong gets much of its power from mainland China, in particular about 70% of the output from Daya Bay's 1888 MWe net nuclear capacity is sent there. A 2014 agreement increases this to 80%. The Hong Kong government plans to close down its coal-fired plants, and by 2020 to get 50% of its power from mainland nuclear (now 23%), 40% from gas locally (now 22%) and 3% from renewables. Another option, with less import dependence, is to increase domestic generation from gas to 60%, and maintain mainland nuclear at 20%.

Hong Kong utility China Light & Power (CLP) has equity in CGN's Daya Bay (25%) power plant, and was until 2013 negotiating a

possible 17% share in Yangjiang. After considering equity in a further CGN nuclear plant, in October 2016 CLP Holdings Ltd successfully bid for a 17% share in Yangjiang Nuclear Power Co Ltd, in response to a CGN general invitation to tender.

Since 1994 Hong Kong has been getting up to one-third of its power from Daya Bay output, and this contract now runs to 2034. According to CLP data, nuclear power cost HK 47 c/kWh in November 2013, compared with 27 cents for coal and 68 cents for gas, which provides the main opportunity to increase supply. CLP supplies about 80% of Hong Kong's power.

Regulation and safety

The National Nuclear Safety Administration (NNSA) was set up in 1984 under the China Atomic Energy Authority (CAEA) and is the licensing and regulatory body which also maintains international agreements regarding safety and non-proliferation safeguards. While it already reported to the State Council directly, in January 2011 the State Council Research Office (SCRO) recommended: "The NNSA should be an entity directly under the State Council Bureau, making it an independent regulatory body with authority," although its international roles would continue to be through the CAEA. It is administered by the Ministry of Environmental Protection (MEP), and the head of NNSA has the rank of Vice Minister of MEP. (The CAEA plans new capacity and approves feasibility studies for new plants – see also SCRO report below.) In relation to the AP1000, the NNSA works closely with the US Nuclear Regulatory Commission.

The NNSA has been working closely with the US Nuclear Regulatory Commission (NRC) in relation to the AP1000, and has adopted some parallel regulations. This, with a high level of IAEA and OECD involvement, gives the NNSA strong international credibility.

The Ministry of Ecology and Environment (MEE) is directly under the State Council and is responsible for radiological monitoring and radioactive waste management and for administration of the NNSA.

A utility proposing a new plant submits feasibility studies to the CAEA, siting proposals to the NNSA and environmental studies to the MEE itself. Nuclear power plant licences issued by the NNSA progress from siting approval, then construction permit, fuel loading permit, to operating licence.

The NNSA is responsible for licensing all nuclear reactors and other facilities, safety inspections and reviews of them, operational regulations, licensing transport of nuclear materials, waste management, and radiation protection including sources and NORM. It licenses staff of nuclear manufacturers through to reactor operators. The NNSA certifies companies for design, manufacture, installation and non-destructive inspection of civilian nuclear safety equipment, and imported equipment. It is responsible for radiological protection and for environmental impact assessment of nuclear projects, in collaboration with the Ministry of Environmental Protection (MEP). The 2003 Law on Prevention and Control of Radioactive Pollution is supplemented by a number of regulations issued since 1986 with the authority of State Council.

In September 2025 China adopted the updated Atomic Energy Law, which entered into force in January 2026.

The NNSA is responsible for safeguards materials accounting throughout the fuel cycle, and communicates with the International Atomic Energy Agency (IAEA) in this regard.

In 2013 the CAEA signed a cooperation agreement with the OECD's Nuclear Energy Agency (NEA), confirming China as a 'key partner' with the OECD. In May 2014 the NNSA signed an agreement on the regulation of nuclear power and radiation safety with the OECD NEA. This is to share experience on the effective regulation and oversight of nuclear safety, as well as best practice in licensing and oversight of civil nuclear facilities.

China has shown unprecedented eagerness to achieve the world's best standards in nuclear safety (as also in civil aviation). It requested and hosted 12 Operational Safety Review Team (OSART) missions from IAEA teams to October 2011, and each plant generally has one external safety review each year, either OSART, WANO peer review, or CNEA peer review (with the Research Institute for Nuclear Power Operations, RINPO). In December 2013 the NNSA with its Japanese and South Korean counterparts agreed to form a network to cooperate on nuclear safety and quickly exchange information in nuclear emergencies. The NNSA is also part of the ASEAN+3 Forum on Nuclear Safety.

Following the Fukushima accident in Japan in March 2011, the government suspended its approval process pending a review of lessons which might be learned from it, particularly regarding siting of reactors with plant layout, and control of radiation release. Safety checks of operating plants were undertaken immediately, and a

review of those under construction was completed in October 2011. Resumption of approvals for further new plants was suspended until a new nuclear safety plan was accepted and State Council approval given in October 2012 (see also Post-Fukushima review below).

Following the Fukushima accident, concern regarding possible river pollution and depletion during droughts (due to evaporative cooling towers) meant many years of delay for the inland AP1000 plants which were due to start construction in 2011.

SCRO report on nuclear investment and safety

In January 2011 a report from the State Council Research Office (SCRO), which makes independent policy recommendations to the State Council on strategic matters, was published. While approving the enormous progress made on many fronts, it cautioned concerning provincial and corporate enthusiasm for new nuclear power plants and said that the 2020 target should be restricted to 70 GWe of new plant actually operating so as to avoid placing undue demand on quality control issues in the supply chain. Another 30 GWe could be under construction. It emphasized that the priority needed to be resolutely on Generation-III technology, notably the AP1000 and derivatives.[3]

SCRO said that China should limit the number of Generation II reactors built, notably CPR-1000. It noted the 100-fold increase in probabilistic safety brought by Generation III, and that future reactor generations would continue the trend.

SCRO drew attention to another factor potentially affecting safety – the nuclear power workforce. While staff can be technically trained in four to eight years, "safety culture takes longer" at the operational level. This issue is magnified in the regulatory regime, where salaries are lower than in industry, and workforce numbers remain relatively low. SCRO said that most countries employ 30-40 regulatory staff per reactor in their fleet, but the National Nuclear Safety Administration (NNSA) had only 1000 staff – a figure that must more than quadruple by 2020. The SCRO recommended: "The NNSA should be an entity directly under the State Council Bureau, making it an independent regulatory body with authority."

Post-Fukushima review

Immediately following the Fukushima accident in March 2011, the State Council announced on March 16 that it would suspend approvals for new nuclear power stations and conduct comprehensive safety checks of all nuclear projects, including those under construction (with an immediate halt required on any not satisfactory). It also suspended work on four approved units due to start construction in 2011*. About 34 reactors were already approved by the central government of which 26 were being built. The Shidaowan HTR, though ready for first concrete, was also delayed. After three months the inspections of operating plants had been completed, and those on plants under construction were completed by October (though construction had continued).

* Fuqing 4-6, Yangjiang 4. Fuqing 4 and Yangjiang 4 started construction late in 2012

In 2012 a new safety plan for nuclear power was approved by State Council, and full incorporation of International Atomic Energy Agency (IAEA) safety standards became explicit. In an unprecedented move to improve the transparency of nuclear regulation the government then formally solicited public comments on its nuclear safety plan which must ensure that no 'serious incident' (INES Level 3) or greater occurs at any reactor. (So far in China no nuclear incident has been over INES level 2.) The plan involved significant expenditure across all of the country's facilities. In October 2012 approvals for new plants recommenced. The nuclear capacity target for 2020 became 58 GWe in operation and 30 GWe under construction.

The State Council in 2012 approved the "12th 5-year Plan for Nuclear Safety and Radioactive Pollution Prevention and Vision for 2020", compiled by the Ministry of Environment. It suggested that China would need to spend RMB 80 billion ($13 billion) on improving nuclear safety at 41 operating and under construction reactors by 2015. "China has multiple types of nuclear reactors, multiple technologies and multiple standards of safety, which makes them hard to manage," it said, adding that the operation and construction of nuclear reactors must improve. The bottom line of safety requirements is that radioactive releases should never cause "unacceptable effects on the environment or the public", and that advanced nuclear technology should "practically eliminate the possibility of release of significant quantities of radioactive substances from nuclear power units" built from 2016.

A series of research and development (R&D) projects was launched by the NEA in February 2012 to improve safety-related technology and the country's emergency response capabilities at indigenous nuclear power plants in the event of an extreme disaster beyond design basis. The 13 R&D projects were conducted by CNNC, CGN and the Institute of Nuclear and New Energy Technology (INET) at Tsinghua University. They include the development of passive emergency power supply and cooling water systems, development of passive containment heat removal systems, developing hydrogen control devices, measures for the prevention and mitigation of used fuel accidents, and analysing the impact of multiple simultaneous external events and response measures. The outcome of this by 2014 was to bring the safety of CPR-1000 reactors up to "internationally recognized levels" required for Generation III reactors.

In 2014, the five major nuclear power utilities including CNNC and CGN signed the *Cooperation Framework Agreement on Mutual Aid for Nuclear Accident Emergency Among Groups*, undertaking to establish and improve the emergency response mechanism and cooperation between adjoining nuclear power plants belonging to different groups, in case of a serious nuclear accident.

International review of NNSA and MEE

In July 2010 a 22-strong IAEA team from 15 countries carried out a two-week Integrated Regulatory Review Service (IRRS) mission to review of China's regulatory framework for nuclear safety. The IAEA made a number of recommendations but said that the review had provided "confidence in the effectiveness of the Chinese safety regulatory system and the future safety of the vast expanding nuclear industry."

In 2016 a further IRRS mission was invited and a 14-strong team from the IAEA reviewed the China situation with many more reactors online and under construction. An interim report said that most of the recommendations made during the 2010 mission had been implemented but that "further work is needed in areas such as managing long-term operation of nuclear power plants and waste management." Also: "The team recommended that China should continue its progress towards adopting the draft Nuclear Safety Act, which sets out fundamental safety principles [and which] should ensure the independence of MEP/NNSA as a regulatory body."

Reactor technology

China has set the following points as key elements of its nuclear energy policy:

- PWRs will be the mainstream but not sole reactor type.
- Nuclear fuel assemblies are fabricated and supplied indigenously.
- Domestic manufacturing of plant and equipment will be maximized, with self-reliance in design and project management.
- International cooperation is nevertheless encouraged.

The technology base for future reactors remains officially undefined, though two designs are currently predominant in construction plans: CAP1000 and Hualong One, after plans for more CPR-1000 units were scaled back post-Fukushima. Beyond them, high-temperature gas-cooled reactors and fast reactors appear to be the main priorities.

A major struggle between the established China National Nuclear Corporation (CNNC) pushing for indigenous technology and the

small but well-connected State Nuclear Power Technology Corp (SNPTC) favouring imported technology was won by SNPTC about 2004. In particular, SNPTC proposed use of indigenized 1000+ MWe plants with advanced third-generation technology, arising from Westinghouse AP1000 designs at Sanmen and Haiyang (see section below on Embarking upon Generation III plants). Westinghouse has agreed to transfer technology to SNPTC over the first four AP1000 units so that SNPTC can build the following ones on its own. In 2014 SNPTC signed a further agreement with Westinghouse to deepen cooperation in relation to AP1000 and CAP1400 technology globally and "establish a mutually beneficial and complementary partnership". In November 2018 successors of the two companies – State Power Investment Corporation (SPIC) and Westinghouse (under new ownership) – signed a further cooperation agreement.

In February 2006, the State Council announced that the large advanced PWR was one of two high priority projects for the next 15 years, depending on "Sino-foreign cooperation, in order to master international advanced technology on nuclear power and develop a Chinese third-generation large PWR".[4] In September 2006, the head of the China Atomic Energy Authority said that he expected large numbers of third-generation PWR reactors derived from foreign technology to be built from about 2016, after experience is gained with the initial AP1000 units.

This trend was given impetus by the reappraisal of safety following the Fukushima accident.

EPR

Two EPR reactors have been built at Taishan, and two more are planned (see section below on Embarking upon Generation III plants). Areva says the reactors are 4590 MWt, with net power 1660 MWe.

In October 2008, Areva and CGN (then: CGNPC) announced establishment of an engineering joint venture as a technology transfer vehicle for development of the EPR and possibly other PWR plants in China and later abroad. The Wecan JV, 55% CGN subsidiary China Nuclear Power Engineering Co. and 45% Areva, was set up in December 2009 and based in Shenzhen, though by mid-2011 the CGN share was held by China Nuclear Power Technology Research Institute (CNPRI), another subsidiary. Overseas projects involving CGN appear now to hold the only

potential for expanding the role of Areva's EPR technology involving China.

AP1000, CAP1000

The Westinghouse AP1000 was to be the main basis of China's move to Generation III technology, and involved a major technology transfer agreement. It is a 1250 MWe gross reactor with two coolant loops. Four AP1000 reactors were completed at Sanmen and Haiyang in 2018, for CNNC and China Power Investment Corp (CPI) respectively. Six more at three sites were firmly planned after them, at Sanmen, Haiyang and Lufeng (for CGN), and at least 30 more were proposed to follow. A State Council Research Office report in January 2011 emphasized that these should have priority over alternative designs such as CPR-1000, and this position strengthened following the Fukushima accident.

The reactors are built from modules fabricated adjacent to each site. The timeline is 50 months from first concrete to fuel loading, then six months to grid connection for the first four units, with this expected to reduce significantly for the following units. In October 2009, SNPTC and CNNC signed an agreement to co-develop and refine the AP1000 design, and this position strengthened following the Fukushima accident. (See also section below on Embarking upon Generation III plants).

CNEA estimated in May 2013 that the construction cost for two AP1000 units at Sanmen would be CNY 40.1 billion ($6.12 billion), or 16,000 yuan/kW installed ($2440/kW), instead of CNY 32.4 billion earlier estimated. This is about 19% higher than the latest estimate for the CPR-1000 (CNY 13,400/kW, $2045/kW), but likely to drop to about that level with series construction and greater localization as envisaged. Grid purchase price is expected to exceed CNY 0.45/kWh at present costs, and drop to the standard CNY 0.43/kWh with series build and reduced capital cost.

Plans for future AP1000 units will now be CAP1000, which is a local standardization of the design, transitional to the CAP1400. It is said to have reduced cost and improved operation and maintenance attributes. The base design, commenced in 2008, is complete, the detailed design, started in April 2010, was due by June 2013*. Early in 2012 SNPTC had organized SNERDI (nuclear island and general designer) and SNPDRI (for conventional island) to localize the design for both inland and coastal sites, for Xianning, Pengze and Taohuajiang. The demonstration CAP1000 plant is under construction at Haiyang (units 3&4). Localization is 80%.

* Differences from the AP1000 include conforming to Chinese design standard GB6429, construction management, supply chain technical requirements, post-Fukushima modification, and module design.

Alstom signed an agreement in July 2013 with Dongfang Electric Corporation for turbine and generator packages for CAP1000 projects to be based on Alstom's Arabelle technology.

CAP1400

Westinghouse announced in 2008 that it was working with SNPTC and Shanghai Nuclear Engineering Research & Design Institute (SNERDI) to develop jointly a passively safe 1400-1500 MWe design from the AP1000/CAP1000, for large-scale deployment, though SNPTC would have full intellectual property rights for any derivatives over 1350 MWe. SNPTC initially called it the Large Advanced Passive PWR Nuclear Power Plant (LPP or APWR). It is one of 16 Key National Projects in China. This development with SNERDI opens the possibility of China itself exporting the new larger units with Westinghouse's cooperation. In April 2016 the IAEA presented SNERDI with its positive evaluation report from the generic reactor safety review process undertaken over nine months. The CAP1400 is also known as 'Guohe No. 1'.

In December 2009, the State Nuclear Plant Demonstration Company – a 55-45% joint venture company by SNPTC and China Huaneng Group – was set up to build and operate an initial demonstration unit of the larger two-loop design, the CAP1400, at Huaneng's Shidaowan site at Rongcheng. The new company signed a set of agreements with SNERDI and the State Nuclear Power Engineering Company (SNPEC) in November 2010 to proceed with the project. It will have 193 fuel assemblies, MOX capability, 50 GWd/t burn-up and improved steam generators. Seismic rating is 300 gal. Dongfang Electric is to design and build the turbine generator under contract to SNPTC.

The basic design was approved by the National Energy Administration (NEA) in January 2014. Site works were completed in April 2014, with final NNSA approvals in September, following a 17-month review. In December 2014 SNPTC said it was ready to pour first concrete but awaited State Council permission, then in April 2015 construction of the turbine hall started. In November 2015 SNPTC said that construction would start in March 2016, the delay being to ensure that the AP1000 primary coolant pump issues were resolved. Construction of unit 1 started in about June

2019 and unit 2 in April 2020. Westinghouse provided technical consulting services to SNPTC for the design. Over 90% of the components are indigenous. Construction cost is expected to be CNY 15,751/kWe ($2450/kWe) and power cost CNY 0.403/kWh for the first unit and dropping to CNY 0.38/kWh (5.9 ¢/kWh) subsequently. A 2014 government figure is CNY 42.3 billion ($6.5 billion) for the first two units. SPIC officially launched the Guohe One project in September 2020 after the first two units were already under construction. Unit 1 entered commercial operation at the end of 2024.

SPIC is promoting the Guohe One brand for export and is discussing potential partnerships with countries including Turkey and South Africa. In May 2016 SNERDI said that the successful IAEA generic reactor safety review would help, "laying a solid foundation for the CAP1400 to participate in international competition at a higher level." After Rongcheng, SNPTC envisaged further CAP1400 projects for Liaoning, Shandong, Fujian, Guangdong and Guangxi provinces.

CAP1400 may be followed by a larger, three-loop CAP1700 design if the passive cooling system can be scaled to that level. Agreements with Westinghouse stipulate that SNPTC will own the intellectual property rights. SNPEC is doing the engineering under a team from SNERDI, the Shandong Electric Power Engineering Consulting Institute (SEPECI), and the State Nuclear Power Equipment Manufacturing Company (SNPEMC), which will make the components.

CNP-1000, also CNP-600, CNP-300, SNP350 (ACP300, ACP600, ACP1000)

CNNC had been working with Westinghouse and Framatome (now Areva) at SNERDI since the early 1990s to develop a Chinese standard three-loop PWR design, the CNP-1000. This is developed from the single-loop Qinshan CNP-300 unit, a 966 MWt, 310 MWe gross, 300 MWe net reactor (uprated by potential 50 MWe in April 2019) with 30-year design operating lifetime (and 20-year extension in 2021). This was then scaled up to the two-loop CNP-600 units, also at Qinshan, with high (60 GWd/t) burn-up, 18-month refuelling cycle and 20 more (but shorter) fuel assemblies than the French-origin M310 units at Daya Bay and Ling Ao.[b] In 1997, the Nuclear Power Institute of China (NPIC) at Chengdu became involved in the reactor design and, early in 2007, SNERDI was reassigned to concentrate on the AP1000 programme.

CNNC was keen to create its own brand of advanced second-generation reactor with full intellectual property rights, and wanted to build two initial CNP-1000 plants at Fangjiashan, adjacent to Qinshan near Shanghai, under the 11th Economic Plan, though the design probably would not have been ready. In early 2007, the CNP-1000 development was put on hold, though this aborted export plans then for two CNP-1000 units to Pakistan. CPR-1000 reactors, designated M310+ by CNNC, were built at Fangjiashan.

Further CNP-600 units were built at Qinshan and Changjiang, Hainan. CNNC says they are free of French intellectual property rights. CNNC is also developing the design to the ACP600 which it calls a third-generation design and expected to be built on Hainan or in the northwest Gansu province and exported.[c] It will have double containment, 18-24 month refuelling cycle, 121 fuel assemblies (as CNP-600), digital I&C, and 60-year operating lifetime, but slightly less power – 605 MWe instead of 650 MWe gross.

In October 2011 CNNC announced that its independently-developed ACP1000 (or CP1000) was entering the engineering design stage, initially for Fuqing units 5&6, with 1100 MWe nominal power and load-following capability. It has 177 fuel assemblies 3.66 m long. In May 2013 CNNC finished a preliminary safety analysis report, and was working on construction design in order to be ready for construction by the end of the year. CNNC expected to start building the first in 2014, at Fuqing, with 85% local content, and the second there in 2015. In April 2013 it announced an export agreement for an ACP1000, for Pakistan. CNNC asserts full intellectual property rights for the CNP series of reactors, which have evolved to the ACP series. However, when the National Energy Administration ordered a rationalization of CNNC's and CGN's 1000 MWe class designs, the ACP1000 morphed into the Hualong One or HPR1000 – though CNNC still sometimes describes its version of this as the ACP1000. Meanwhile the IAEA approved the ACP1000 design in its generic reactor safety review process in December 2014, though the IAEA points out that this does not "constitute any kind of design certification."

Two new 300 MWe CNP-300 PWR units have been built at Chasma in Pakistan by the China Zhongyuan Engineering Corporation. They are similar to those already commissioned at that site in 2000 and 2011, and similar to Qinshan 1 – China's first indigenously-designed (by SNERDI) nuclear power plant.

The **SNP350** is SNERDI's development of the CNP-300, upgraded in many respects to meet latest performance, economy, and safety

requirements. It is 1035 MWt, 350 MWe gross, with a design operating lifetime of 60 years and digital I&C systems. In series its cost is expected to be $4000/kW. If built by CNNC it may be designated CNP350.

CNNC was seeking to sell the CNP-300 to Belarus and in Africa, and these will probably now become ACP300 or SNP/CNP350.

ACP100 small modular PWR

A 'key project' in the 12th Five-Year Plan was CNNC's multi-purpose small modular reactor, the ACP100, or LingLong One. The preliminary design was completed in 2014, based on the larger ACP (and CNP) units and a preliminary safety analysis report (PSAR) has been completed. The basic design was completed early in 2016, and CNNC said that the NDRC approved the ACP100S design for marine use late in 2016. The design took four years and was funded to CNY 700 million. In April 2016 the IAEA presented CNNC with its report from the generic reactor safety review process undertaken over ten months. In 2016 CNNC subsidiary China Nuclear Power Engineering Corp (CNPE) submitted an expression of interest to the UK government based on its ACP100+ design. By 2030 CNNC expects the ACP100 to be standardized as an advanced SMR, with large-scale construction.

It has passive safety features and will normally be installed underground. Seismic tolerance is 300 Gal. It has 57 fuel assemblies 2.15m tall and integral steam generators (320°C input), so that the whole steam supply system is produced and shipped as a single reactor module. It has passive cooling for decay heat removal. Its 385 MWt produces about 125 MWe, and power plants comprising two to six of these are envisaged, with 60-year design life and 24-month refuelling. Or each module can supply 1000 GJ/hr, giving 12,000 m^3/day desalination (with MED). Industrial and district heat uses are also envisaged, as is floating nuclear power plant (FNPP) application. See further in Floating nuclear power plants section below.

CNNC New Energy Corporation (CNNC-CNEC), a joint venture of CNNC (51%) and China Guodian Corp, was planning to build two ACP100 units in Putian county, Zhangzhou city, at the south of Fujian province, near Xiamen and not far from Fuqing, as a demonstration plant. This would be the CNY 5 billion ($788 million) phase 1 of a larger project. Construction time was expected to be 36-40 months for the two ACP100 units. CNNC applied for NDRC approval, but early in 2017 the site for first ACP100 units was

changed to Changjiang, on Hainan, with a larger reactor to be built at Putian. In May 2017 CNNC said it was ready to build the first unit at Changjiang. Preliminary work began in July 2019 and full construction has begun, but no official announcement has been made. 58 months construction is expected.

A second proposal was approved for two further CNNC-CNEC units at Zhangzhou-Gulei, but this is suspended by local government opposition.

The Linglong One Demonstration Project* involves a joint venture of three companies: CNNC as owner and operator, Nuclear Power Institute of China (NPIC) as the reactor designer, and China Nuclear Power Engineering Group being responsible for plant construction. CNNC-CNEC signed a second ACP100 agreement with Hengfeng county, Shangrao city in Jiangxi province, and a third with Ningdu county, Ganzhou city in Jiangxi province in July 2013 for another ACP100 project costing CNY 16 billion. Further inland units are planned in Hunan and Jilin provinces, and CNNC has signed ACP100 development agreements also with Zhejiang and Heilongjiang provinces. Export potential is considered high, with full intellectual property rights. CNNC-CNEC will construct major parts of the reactors in Bashan, Jilin province. For the demonstration plant, the reactor vessel is from Shanghai Boiler Co, the steam generators from a CNNC subsidiary and other reactor internals from Dongfang.

* Hainan Changjiang Multi-Purpose Small Modular Reactor Technical Demonstration Project is its full name.

In August 2023 CNNC announced the installation of the integrated reactor core module for the Linglong One Demonstration Project. In September 2025 it announced that containment integrity testing had been completed, and in October cold functional tests were completed. In December 2025 the steam turbine test was completed, with commissioning expected in the first half of 2026.

CAP200, CAP150, CAP50 small modular PWRs

These have SPIC/SNPTC provenance, being developed from the CAP1000 in parallel with the CAP1400 by SNERDI, using proven fuel and core design. SNPTC is focused on the CAP200 and CAP50, the latter for floating nuclear power plants and desalination.

CAP200 or **LandStar-V** from SNERDI is 660 MWt/220 MWe and has two external steam generators (301°C). It is pitched to replace

coal plants and supply district heating, and has a design operating lifetime of 60 years. With 24-month refuelling, burn-up of 42 GWd/t is expected. The 89 fuel assemblies are the same as for the CAP1400 but shorter. It has both active and passive core cooling systems, with natural circulation effective for up to 20% power. The CAP200 will be installed in a 32 m deep underground caisson structure with a seismic design basis of 600 Gal, even in soft ground. In 2017 the first-of-a-kind cost was estimated at $5000/kW and $160/MWh, dropping to $4000/kW in series.

The earlier **CAP150** design is 450 MWt/150 MWe. The design has eight integral steam generators (295°C), and is claimed to have "a more simplified system and more safety than current third generation reactors." It is pitched for remote electricity supply and district heating, with three-year refuelling and design operating lifetime of 80 years. It has both active and passive cooling and in an accident scenario, no operator intervention is required for seven days. The seismic design basis is 300 Gal. In mid-2013 SNPTC quoted approximately $5000/kW capital cost and 9 c/kWh for the CAP150, so significantly more than the CAP1400.

A related SNERDI project is a reactor for floating nuclear power plants. This is to use the **CAP50** of about 200 MWt and relatively low-temperature (250°C), so only about 40 MWe with two external steam generators and five-year refuelling. SPIC has also presented the OceanStar-V version of the CAP200, on a barge, as a floating nuclear power plant.

CPR-1000, M310+, ACPR1000

The CPR-1000 is a significantly upgraded version of the 900 MWe-class French M310 three-loop technology imported for the Daya Bay nuclear power plant in the 1980s and also built at Ling Ao. Known as the 'improved Chinese PWR' and designated Generation II+, it features digital instrumentation and control and a design life of 60 years. Its 157 fuel assemblies (3.66 m – 12 feet – long) have calculated core melt frequency of $1x10^{-5}$ and a release probability an order of magnitude lower than this.

Standard construction time is 52 months, and the claimed unit cost was under CNY 10,000 (US$ 1600) per kilowatt, though 2013 estimates put it at about $2300/kW domestically. With a capacity of 1080-1089 MWe gross (usually about 1020 MWe net), Ling Ao Phase II is the first plant to be designated as the CPR-1000 design. It cost CNY 28.5 billion, hence CNY 14,150/kW. (The Hongyanhe units, in colder water, are 1119 MWe gross, 1061 MWe net.) The

CPR-1000 was being widely and quickly deployed for domestic use, with 57 likely to be built, as of end of 2010. Following the Fukushima accident, no further approvals were made.

China Guangdong Nuclear Power Corporation (CGN) led the development of the CPR-1000 and established a nearly complete domestic supply chain. However, Areva retains intellectual property rights, which constrains overseas sales since the Chinese would need agreement from Areva on a case-by-case basis.

The eight CNNC-built units derived from the M310 (Fangjiashan 1&2, Fuqing 1-4 and Tianwan 5&6) are designated M310+.

The **Advanced CPR** – **ACPR1000** – with full Chinese intellectual property rights, was launched by CGNPC in November 2011 with some fanfare regarding its safety attributes, which comply with international requirements. CGN has been in cooperation with Dongfang Electric, Shanghai Electric, Harbin Electric, China First Heavy Industries, China Erzhong and other companies since 2009 to develop the ACPR1000, a three-loop unit with double containment and core-catcher. Yangjiang 3&4 are an intermediate CPR-1000+, with some design modifications.

CGN made the ACPR1000 available for local build on schedule from 2013 with the first at Yangjiang, units 5&6, followed by Hongyanhe 5&6 and Lufeng 1&2 (now to be CAP1000). In September 2012 Fangchenggang 3&4 was identified as the demonstration project, with construction start at the end of 2014, but meanwhile construction started on Yangjiang 5&6, which CGN said had evolved to be an ACPR1000 design. Fangchenggang 3 was to be the reference plant for CGN's bid to build the Sinop plant in Turkey. Overnight construction cost is expected to be $2500/kW.

A further development, ACPR1000+, was envisaged for export, from 2014, but was abandoned with the rationalization to Hualong One described below, CGN's version of which is the HPR1000. It was to have a 60-year life and 300 Gal seismic capability. The conceptual design gained IAEA approval through its generic reactor safety review process in May 2013, though the IAEA points out that this does not "constitute any kind of design certification".

In January 2012, CGN with Areva and EdF agreed on a partnership to develop a Generation III reactor based on the CPR-1000, the ACE1000 (Areva-CGN-EdF1000). Mitsubishi Heavy Industries, already designing the similar-size Atmea1 with Areva, said it will not be involved. The ACE1000 concept is reported to be evolutionary, with single but reinforced containment, active safety systems with three 100% loops (so can do maintenance on one of

them outside outages). No more has been heard of this plan and it has evidently been overtaken by Hualong One.

ACPR small modular PWRs

Not to be outdone by CNNC in the small modular field, CGN has two small ACPR designs: an ACPR100 and an ACPR50S, both with passive cooling for decay heat and 60-year design life. Both have standard type fuel assemblies and fuel enriched to <5% with burnable poison giving 30 month refueling. The **ACPR100** is an integral PWR, 450 MWt, 140 MWe, having 69 fuel assemblies. Reactor pressure vessel is 17m high and 4.4 m inside diameter, operating at 310°C. It is designed as a module in larger plant and would be installed underground.

The offshore **ACPR50S** is 200 MWt, 60 MWe with 37 fuel assemblies and four external steam generators. Reactor pressure vessel is 7.4m high and 2.5 m inside diameter, operating at 310°C. It is designed for mounting on a barge as floating nuclear power plant (FNPP) or possible submersible. The applications for these are similar to those for the ACP100. CGN announced in January 2016 that development of its reactor design has been approved by the NDRC as part of the 13th Five-Year Plan for innovative energy technologies. In November 2016 it announced that it had contracted with Dongfang Electric Corporation for the pressure vessel of a demonstration ACPR50S unit, which it said amounted to start of construction. See further in Floating nuclear power plants section below.

Hualong – HPR1000

Since 2011 several rounds of negotiations between CNNC and CGN grappled with the task of "merging" the **ACP1000** and **ACPR1000+** designs as ordered by the National Energy Administration (NEA) while allowing for some differences, with impetus given by the regulator. Both are three-loop designs based to some extent on the French M310, but the cores are very different: one (ACP1000) evolved from the established CPR1000, has 177 fuel assemblies 3.66 m (12 feet) long, while the ACPR1000, with four units at Hongyanhe and Yangjiang, has 157 assemblies 4.3 m (14 feet) long, so physically merging the basic designs was impractical, and in the event the ACP1000 design prevailed in the rationalization. Some features of the ACPR1000 are incorporated into the Hualong One design, at least in the CGN version, which it

calls the HPR1000. The Hualong One requires about 50% more steel and concrete than the CPR1000 and costs about 31% more.

The Hualong One or HPR1000 thus has 177 fuel assemblies 3.66 m long, 18-24 month refuelling interval, equilibrium fuel load will be 72 assemblies with 4.45% enriched fuel. It has three coolant loops, double containment and active safety systems with some passive elements, and a 60-year design life. The passive systems are able to operate for 72 hours with a sufficient inventory of storage water and dedicated batteries. The CGN version delivers 3150 MWt, 1150 MWe gross, 1092 MWe net, while CNNC quotes 3050 MWt, 1170 MWe gross, 1090 MWe net. Average burn-up is 45,000 MWd/tU, thermal efficiency 36%. Seismic tolerance is 300 Gal. Instrumentation and control systems will be from Areva-Siemens, but overall 90% will be indigenous components. Target cost in China is $2800-3000/kWe, though recent estimates mention $3500/kW. CGN said in November 2015 that the series construction cost would be CNY 17,000/kWe (then $2650/kWe), compared with CNY 13,000/kWe for second-generation reactors.

The CNNC and CGN versions are very similar but not identical; they have slightly different safety systems, with CNNC emphasizing more passive safety under AP1000 influence with increased containment volume and two active safety trains, and CGN with French influence having three active safety trains. Also each organization will maintain much of its own supply chain. The new design has been variously called Hualong 1000 or HL1000 by CGN, ACP1000 by CNNC, and generically Hualong One, or HPR1000 – Hualong Pressurized Reactor 1000.

The first units are Fangchenggang 3&4 (CGN) and Fuqing 5&6 (CNNC). The 'rationalization' was helped by greater commonality in ownership of the two companies as set out in September 2012 though still not implemented a year later. Fuqing 5 started construction in May 2015, and it then emerged that the primary coolant pumps were behind schedule. Harbin Electric Company (HEC) formed a special team with CNNC to address the issue. The CGN version of the Hualong One, Fangchenggang 3&4, was to be the reference plant for the UK's (now cancelled) Bradwell B. Fangchenggang 3 construction started in December 2015, followed by unit 4 a year later. The two units were connected to the grid in January 2023 and April 2024 respectively.

CNNC and CGN in December 2015 formed a 50-50 joint venture company – Hualong International Nuclear Power Technology Co – to market the Hualong One. The company is "committed to the continued integration and development of Hualong One as an

independent third generation nuclear power technology, with the unified management of the Hualong brand, intellectual property and other related assets at home and abroad." In some respects the Hualong was originally conceived as a reactor for export but in reality and given the difficulties of the AP1000 in recent years, notably on canned motor pumps, it has emerged as a competitor on national soil and an alternative to the CAP1000.

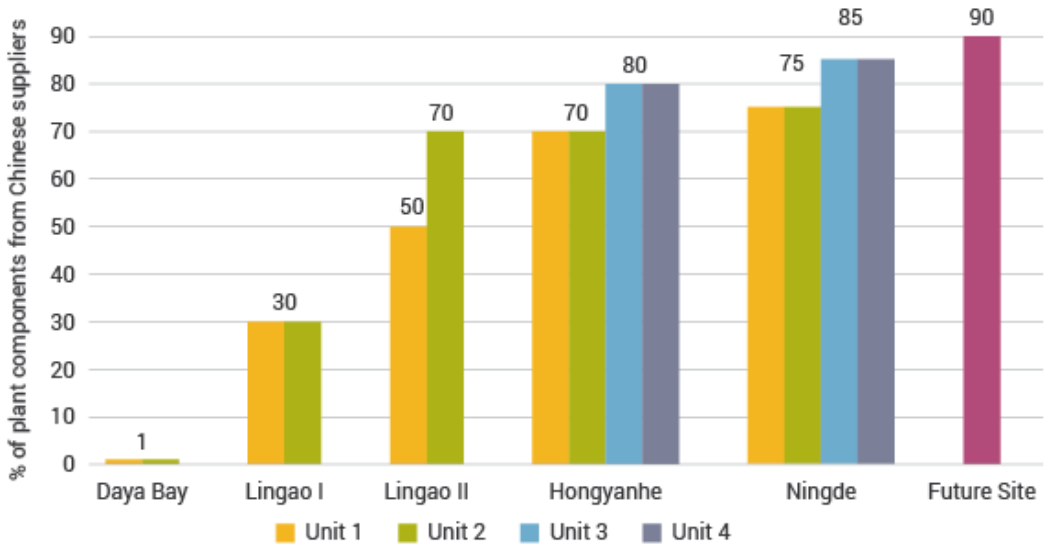
It appears that the CNNC 'Hualong One' version will be the main domestic model built with the aim of lowering the price of the reactor to equip the national fleet cheaply. After Fuqing 5&6 CNNC is building the next at Zhangzhou in Fujian province, previously an AP1000 project. This reactor design will be dedicated to the domestic market and some international markets such as Pakistan, Argentina, and possibly Eastern Europe. It is under construction in Pakistan.

The Hualong One promoted on the international market is called the HPR1000. It is based on the CGN version, with Fangchenggang as reference, and CGN is also building it as Ningde 5&6. In October 2015 CGN submitted the HPR1000 for certification of compliance with the European Utility Requirements (EUR). In March 2016 CGN signed an agreement with Skoda Praha to facilitate this. In July 2017 ten Chinese supply chain companies signed up for ongoing high-level coordination under CGN leadership for enhancing the reactor's design, supply chain, manufacture and cost.

General Nuclear Systems (GNS), the joint venture of EDF Energy and CGN formed to progress the UK generic design approval (GDA) wrote to the UK government in October 2016 saying it was ready to start the GDA process. CGN plans to increase its efforts to sell the reactor in Central and Eastern Europe, Africa, Southeast Asia and elsewhere.

In April 2021 CNNC announced an upgraded and simplified version of the HPR1000 called Hualong Two, with plans then to begin building the first by 2024. It estimates that construction costs would fall to CNY 13,000 ($2000) per kilowatt from the current Hualong One CNY 17,000/kWe ($2600), and construction time will reduce from five years to four.

CGNPC Progressive Localisation of CPR-1000



Source: World Nuclear Association

VVER

Russia's Atomstroyexport was general contractor and equipment provider for the Tianwan 1&2 AES-91 power plants using the V-428 version of the well-proven VVER-1000 reactor of 1060 MWe capacity. The reactors incorporate Finnish safety features and Siemens-Areva instrumentation and control (I&C) systems. Russia's Energoatom is responsible for maintenance from 2009. Tianwan units 3&4 are both the same version of the VVER-1000 reactor with Areva I&C systems.

In June 2018 Rosatom confirmed that it had signed a contract to build four VVER-1200 units in China – two at Tianwan (units 7&8), and two at Xudapu (units 3&4). The contract is reportedly valued at more than CNY 20 billion. China will provide finance, with Russia responsible for the nuclear island only – indicating significant localization.

Candu

From 1998 Atomic Energy of Canada Ltd (AECL) built the two-unit Qinshan Phase III plant on schedule and under budget and estimates that it could be replicated for 25% lower cost. Any replication would be on the basis of involving local engineering teams, not on a turnkey basis, but the technology is now well understood and the decades-old Candu-6 design would likely pose fewer problems for technology transfer than state-of-the-art third-generation designs from Westinghouse and Areva. (The Candu-6

units at Wolsong 2-4 in Korea had substantial local content, reaching 75% localization with unit 4.)

In September 2005, AECL signed a technology development agreement with CNNC which opened the possibility of it supplying further Candu-6 reactors and undertaking fuel cycle developments based on them. This agreement with CNNC was passed to its subsidiary, the Nuclear Power Institute of China (NPIC). From 2008 it has focused on joint development of the Advanced Fuel Cycle Candu Reactor (AFCR).

Advanced Fuel Candu Reactor

In September 2016 an agreement among SNC-Lavalin, CNNC and Shanghai Electric Group (SEC) was to set up a joint venture in mid-2017 to develop, market and build the Advanced Fuel Candu Reactor (AFCR), which will utilize used fuel from other reactors. Two design centres are envisaged, in China and Canada, to complete the AFCR technology. This could lead to construction of two AFCR units in China.

High temperature gas-cooled reactors: HTR-PM, HTR-PM 600

In February 2006, the State Council announced that the small high-temperature gas-cooled reactor (HTR) was the second of two high priority National Major Science & Technology Projects for the next 15 years. This aims at exploring co-generation options in the near-term and producing hydrogen in the long term.

The small HTR-PM (HTR Pebble-bed Modular) units with pebble bed fuel and helium coolant were to be 200 MWe reactors, similar to that then being developed in South Africa, but plans have evolved to make them twin 250 MWt (105 MWe equivalent) reactors so that they can retain the same core configuration as the prototype HTR-10. The twin units, each with a single steam generator, will drive a single 210 MWe steam turbine. Core height is 11 metres in a 25 m high, 5.7 m diameter reactor vessel, with helium at 750°C and 7 MPa producing steam at 566 °C. The engineering of the key structures, systems, and components is based on Chinese capabilities, though they include completely new technical features. Thermal efficiency of 40%, localization 75%, and 50-month construction for the first twin-unit HTR-PM was envisaged.

Each reactor will be loaded with 245,318 fuel elements, each 60 mm diameter and containing 7 g of fuel enriched to 8.5%. (The HTR-10 has 27,000 fuel elements but 17% enrichment.) Average burn-up is 90 MWd/tU. Some graphite moderator pebbles the same size are included. Pebbles are released into the top of the core one by one with the reactor operating. They are correspondingly removed from the bottom, broken ones are separated, the burn-up is measured, and spent fuel elements are screened out and transferred to storage. Reactivity control is primarily by 24 control rods, secondarily by small absorber spheres dropped by gravity through six channels alongside the control rods in the graphite side reflector.

China Huaneng Group (CHNG) is the lead organization in the consortium to build the demonstration Shidaowan HTR-PM with China Nuclear Engineering & Construction Group (CNEC/CNECC)* and Tsinghua University's INET, which is the R&D leader. Chinergy Co., a joint venture of Tsinghua and CNEC, is the main contractor for the nuclear island. Following the agreement on HTR industrialization cooperation between CNEC and Tsinghua University in 2003, the two parties signed a further agreement on commercialization of the HTR in March 2014. CNEC is responsible for the HTR technical implementation, and becomes the main investor of HTR commercial promotion at home and abroad.

* In January 2018, CNNC became a controlling shareholder in CNEC/CNECC, and the two organizations formally merged. For full details see Appendix 1 to this page, Government Structure and Ownership.

Thus the initial demonstration HTR-PM at Shidaowan (Shidaowan HTR-PM 1), near Weihai city, will pave the way for commercial versions with three modules each 2x250 MWt but feeding one turbine, total 655 MWe. Thermal efficiency is 43.7%. The first of the twin reactors of the demonstration HTR-PM achieved first criticality in September and the second in November 2021. The plant entered commercial operation in December 2023.

The HTR-PM 600 reactor units are likely to be built in pairs. INET expects the cost to be 15-20% more than a 600 MWe PWR and the first one to be 75% of the $6000 per kW cost of the small HTR-PM, subsequent ones about 50%. The 655 MWe size and 556 °C steam temperature mean that they can readily replace steam systems in many existing coal-fired plants.

The main HTR promotion is of the 600 MWe version, the HTR-PM 600. In April 2015 CNEC announced that its proposal for two

commercial 600 MWe HTRs at Ruijin city in Jiangxi province had passed an initial feasibility review. At that time, construction was expected to start in 2017. No major utility had been named at that stage, but in July 2016 CNEC signed an agreement with CGN to set up a joint venture led by CNEC to develop HTRs domestically and overseas. It then signed a cooperation agreement with China First Heavy Industries (CFHI or YiZhong) to support the supply chain.

CNEC has done a feasibility study and commenced preliminary work on a commercial-scale HTR-PM600 at Wan'an in Fujian province, close to Ningde, according to CNEC subsidiary China Nuclear Industry Huachen Construction Co. This appears to be instead of Ruijin which is inland. Other feasibility studies for the HTR-PM600 reported by Chinergy in September 2017 are at Sanmen in Zhejiang, Xiapu in Fujian, and Bai'an in Gunagdong.

In 2016 CNEC signed agreements to develop HTRs in Saudi Arabia and Indonesia.

District heating reactors

The government has made clean-energy heating a priority, and in 2017 it issued guidance on this for northern China. The National Energy Administration's five-year plan covering 2017-2021 highlighted clean heating technology.

CNNC, CGN and SPIC have announced concepts for low-temperature district heating reactors, all to be installed well-below ground level. Development of these acknowledges the role of heating in air pollution, particularly $PM_{2.5}$ particulates, which are reported to be more than ten times higher in winter. However, the need for district heating is seasonal, and some use for the plant for the rest of the year is needed.

CGN's NHR200-I is a low-temperature district heating reactor of 200 MWt with convection circulation at 2.5 MPa in primary circuit pressure to produce steam at 127°C. Used fuel is stored around the core in the pressure vessel. Its design is described by CGN as "mature", having passed NNSA review in the 1990s. It has been developed by the Institute of Nuclear Energy Technology (INET) at Tsinghua University from the 5 MW NHR-5 prototype which started up in 1989 and heated the INET campus for three winters. In February 2018 it was announced that CGN and Tsinghua University were carrying out a feasibility study on constructing China's first district heating nuclear plant using the NHR-200 design at Daqing city in Heilongjiang province and Shenyang in

Liaoning province. The NHR200-II with design and verification tests concluded in 2016 operates at 8 MPa primary circuit pressure to produce steam at over 200°C and can also be used for power generation, seawater desalination or heat for mineral processing.

SPIC's LandStar-I is an integral pressure vessel reactor of 200 MWt with convection circulation at 9 MPa producing hot water at 110°C. Pre-feasibility studies suggested first commissioning in 2022. At SPIC's Jiamusi demonstration project in Heilongjiang province, the two 200 MW LS-I reactors were to provide steam to a biomass plant in summer. The National Nuclear Safety Administration (NNSA) approved a preliminary environmental site assessment in August 2020 and they were expected to be in operation in 2024, but did not proceed. LandStar-V is a 600 MWt compact reactor with two external steam generators for power generation or industrial process heat. OceanStar versions are for floating nuclear power plants.

CNNC's Yanlong District Heating Reactor-400 (DHR-400) is a low-temperature 400 MWt pool-type reactor designed to provide heat at 90°C for up to 200,000 three-bedroom apartments. The reactor prototype at CIAE achieved 168 hours of continuous heat supply in November 2017 – seen by its developers, CNNC, as the first major step towards commercialization of the design.

SPIC's Haiyang power plant commenced supplying district heat on a commercial basis in November 2020, and by late 2021 a figure of 6.6 million tonnes of coal (13 PJ) being replaced annually was quoted. A major district heating offtake is planned for the CAP1400 under construction in Shandong, and for power plants at Qinshan, Tianwan and Hongyanhe.

Fast neutron reactors

Longer-term, fast neutron reactors (FNRs) are seen as the main technology, and CNNC expects the FNR to become predominant by mid-century. A 65 MWt fast neutron reactor – the Chinese Experimental Fast Reactor (CEFR) – near Beijing achieved criticality in July 2010, and was grid-connected a year later (but is omitted from the operating table below as being minor and experimental).[5] Based on this, a 600 MWe pre-conceptual design was developed, known as the Chinese Demonstration Fast Reactor (CDFR) project 1 (and sometimes CMFR – Chinese Model Fast Reactor). The first CFR600 unit began construction in December 2017 at Xiapu in Fujian province, and began commissioning in 2023. Construction of the second unit began in December 2020. The CFR600 units are

intended to be followed by a CFR1000 for commercial operation from 2030, according to the China Institute of Atomic Energy. The preliminary design of the CFR1000 was completed in July 2025. Xiapu in Fujian province is also reported to be the site for the prototype so-called travelling-wave reactor TWR-P.

In addition to CDFR project 1, in October 2009, an agreement with Russia confirmed China's intention to opt for the BN-800 technology as CDFR project 2. The first two in China were planned to start construction in 2013 at Sanming, Fujian province (see section below on Sanming). However, despite promotion by NIAEP-Atomstroyexport the project did not proceed.

See also Fast neutron reactors subsection in the R&D section of the information paper on *China's Nuclear Fuel Cycle*.

Thorium molten salt reactor

A 2 MWt experimental thorium molten salt reactor (TMSR-LF1), developed by the Shanghai Institute of Applied Physics, achieved first criticality in October 2023 and reached full power in June 2024. In October 2025 the reactor achieved the first thorium-to-uranium nuclear fuel conversion in a molten salt reactor, with protactinium-233 detected after ten days of full-power operation. China aims to build a 100 MWt demonstration thorium reactor by 2035.

Floating nuclear power plants

In May 2014 the China Atomic Energy Authority (CAEA) signed an agreement with Rosatom to cooperate in construction of floating nuclear cogeneration plants (FNPPs) for China offshore islands. These would be built in China but be based on Russian technology, and possibly using Russian KLT-40S reactors – Russia's TVEL anticipated providing fuel for them. In July 2014 Rusatom Overseas signed a further agreement, this time with CNNC New Energy, for the joint development of FNPPs – both barge-mounted and self-propelled – from 2019.

Since then however, both CNNC and CGN have announced proposals for indigenous FNPPs based on their respective small reactor designs. In February 2017 the State Administration of Science, Technology and Industry for National Defence (SASTIND) and the National Atomic Energy Agency (NEA) said that offshore FNPPs were included in the 13th five-year plans for national defence science and development of the nuclear industry, to

establish relevant standards and undertake key technology research.

In July 2016 CNNC signed an agreement with China State Shipbuilding Corporation (CSSC) to develop nuclear-powered icebreakers and maritime nuclear power platforms. In August 2017 CNNC set up a company to develop, construct, operate and manage FNPPs and other vessels. China Marine Nuclear Power Development Co (CMNPD) is a joint venture among CNNP (51%), Shanghai Guosheng (Group) (20%), Jiangnan Shipbuilding (Group) (10%), Shanghai Electric Group (10%), and Zhejiang Zheneng Electric Power (9%). Jiangnan Shipbuilding near Shanghai is a subsidiary of CSSC. CNNC subsidiary China National Nuclear Power (CNNP) said that establishing the new company, capitalized at CNY 1 billion, was in accordance with the national 'Belt and Road' initiative, to help exploitation of marine resources. Desalination would be part of the role.

In June 2018 CMNPD through CNNC published a tender for construction of a multi-purpose nuclear-powered icebreaker. The 'Linglong One' ACP-100S reactor is envisaged for power.

CGN announced in January 2016 that development of its ACPR50S reactor design has been approved by the NDRC as part of the 13th Five-Year Plan for innovative energy technologies.

CGN then signed an agreement with China National Offshore Oil Corporation (CNOOC) apparently to provide power for offshore oil and gas exploration and production, and to “push forward the organic integration of the offshore oil industry and the nuclear power industry,” according to CNOOC. Soon after, it signed an agreement with China’s largest ship design institute, China Shipbuilding Industry Corporation (CSIC) to "jointly promote cooperation on nuclear power offshore platform projects." In April 2016 CSIC confirmed that in collaboration with CGN it would build the first FNPP at the Bohai Heavy Industries shipyard in Liaoning, supplying power and desalination. (Bohai Shipyard at Huladao is the sole builder of nuclear submarines.)

In October 2018 CSIC presented a concept of a FNPP 163m x 29m displacing 29,800 tonnes with twin reactors providing 200MW total and 50 and 100 MWe, designed by China Shipbuilding Heavy Industry Co Ltd. CSIC has an in-house HHP25 nuclear reactor design derived from a submarine power plant. In early 2019 CGN tendered for construction of an "experimental ship platform" 152 metres long, 30 metres wide, with a displacement of 30,000 tonnes,

dynamic positioning capabilities, and powered by two 25 MW reactors.

In November 2016 CGN announced that it had contracted with Dongfang Electric Corporation for the pressure vessel of a demonstration ACPR50S unit, which it said amounted to start of construction. There are plans for 20 more. CSIC has already designed both a FNPP concept and a submersible nuclear power plant concept.

CNNC announced in October 2015 that its subsidiary, the Nuclear Power Institute of China (NPIC), signed an agreement with UK-based Lloyd's Register to support the development of a floating nuclear power plant using the ACP100S reactor, a marine version of the ACP100. Following approval by the NDRC as part of the 13th Five-Year Plan for innovative energy technologies, CNNC said it was planning to start building its ACP100S demonstration floating nuclear plant late in 2016, for 2019 operation. In July 2016 CNNC signed an agreement with CSIC to give effect to its plans.

Embarking upon Generation III plants

In September 2004, the State Council approved plans for two units at Sanmen, followed by six units at Yangjiang (two to start with), these to be 1000 or 1500 MWe reactors pioneering Generation III nuclear technology from overseas. The Sanmen (in Zhejiang province) and Yangjiang (in Guangdong province) reactors were subject to an open bidding process for third-generation designs, with contracts to be awarded in mid-2006 – in the event, mid-2007 – putting them clearly into the 11th Five Year Plan.

Bidding process

This open bidding process underlined the extent to which China is making itself part of the world nuclear industry, and yet at first remaining somewhat ambivalent about that.

Three bids were received for the four Sanmen and Yangjiang reactors: from Westinghouse (AP1000 reactors), Areva (EPR) and Atomstroyexport (VVER-1000 model V-392). The State Nuclear Power Technology Corporation (SNPTC), directly under China's State Council, was in charge of technology selection for new plants being bid from overseas.

Some 200 experts spent over a year evaluating Generation III designs and in September 2006 most of the 34 assigned to decide voted for the AP1000. The key factors in choosing this were passive design, simplified safety system, modular construction giving more rapid build and better cost control, and smaller components allowing more ready localization.

The USA, French and Russian governments were reported to be giving firm support as finance and support arrangements were put in place. The US Export-Import bank approved $5 billion in loan guarantees for the Westinghouse bid, and the French Coface company was expected similarly to finance Areva for its bid. The US Nuclear Regulatory Commission gave approval for Westinghouse to export equipment and engineering services as well as the initial fuel load and one replacement for the four units. Bids for both two-unit plants were received in Beijing on behalf of the two customers: China General Nuclear Power Co (CGN) for Yangjiang, and China National Nuclear Corporation (CNNC) for Sanmen. Bids were for the nuclear portion of each plant only, the turbine tenders to be called for subsequently.

Bids were assessed on level of technology, the degree to which it was proven, price, local content, and technology transfer – which apparently became the major factor. Areva and Westinghouse were short-listed. However, the decision on reactor type was delayed, and came under review at the highest political level, with CNNC evidently pushing for the use of indigenous second-generation designs for both sites.

In December 2006, 22 months after the bids were submitted and after several revisions to them, the Westinghouse AP1000 reactor design was confirmed for the four units – two each at Sanmen and Yangjiang. Early in 2007, the two units planned for the Yangjiang site were switched to Haiyang in the more northerly Shandong province, making way for two EPR units Areva was in negotiations to build at Yangjiang. Later in 2007, plans for the EPRs under consideration for Yangjiang were transferred to another Guangdong site – Taishan – since there was pressure to build a lot of capacity quickly at Yangjiang.

Sanmen 1&2 and Haiyang 1&2

A framework agreement was signed at the end of February 2007 between Westinghouse and SNPTC specifying Haiyang and Sanmen for the four AP1000 units. In July 2007, Westinghouse, along with consortium partner Shaw, signed the contracts with

SNPTC, Sanmen Nuclear Power Company (51% owned by CNNC), Shandong Nuclear Power Company (65% owned by CPI, for Haiyang) and China National Technical Import & Export Corporation (CNTIC) for four AP1000 reactors. Specific terms were not disclosed but the figure of $5.3 billion for the first two was widely quoted. In December 2007 the Chinese and US governments signed the intergovernmental agreement for the construction of AP1000 projects in China and technology transfer.

Sanmen site works commenced in February 2008 and full construction on Sanmen 1 – the world's first AP1000 unit – officially commenced on 19 April 2009. The reactor was initially expected to begin operation at the end of 2015 with the second less than one year later. First fuel was loaded into Sanmen 1 in April 2018, the unit was grid connected in June 2018 and entered commercial operation in September 2018. First fuel was loaded into Sanmen 2 in July 2018, the unit was grid connected in August 2018, and entered commercial operation in November 2018.

First concrete at Haiyang 1 was in September 2009, and the pressure vessel was installed in January 2012. First fuel was loaded into Haiyang 1 in June 2018, the unit was grid connected in August 2018, and entered commercial operation in October 2018. First fuel was loaded into Haiyang 2 in August 2018, the unit was grid connected in October 2018, and entered commercial operation in January 2019. China Nuclear Engineering & Construction Group (CNEC/CNECC) is building both the Sanmen and Haiyang plants. In December 2019, the two-unit Haiyang plant began providing heat to over 700,000 m^2 of housing as part of the first phase of the Shandong Haiyang Nuclear Energy Heating Project. In May 2020 the project received approval to operate as the country's first large-sale cogeneration heating project, allowing it to provide heating for 30,000,000 m^2 of housing.

The future of Haiyang, as envisaged by CNNC

Construction was slower than planned, the main problem being with US-made main coolant pumps (four in each reactor). These worked well in normal operation but had insufficient inertia to continue long enough for full passive safety effectiveness when not powered. Following successful tests,* the modified pumps were shipped from Curtiss-Wright to Sanmen, and the fourth was installed in unit 1 in March 2016. This is also a milestone enabling modified pumps to be supplied to seven other AP1000 units under construction in both China and the USA.

* The final performance testing verified successful pump operation during a full range of AP1000 plant operating conditions for more than 1,600 total hours, including more than 600 start-and-stop cycles. Extensive reviews and detailed post-test inspections by the NNSA confirmed performance as designed during the final testing. Curtiss-Wright conducted the testing and inspection at its Electro-Mechanical Division (EMD) facility in Cheswick, Pennsylvania, USA. SNPTC said: "This success will further accelerate China AP1000 project construction progress as well as boost advanced passive Generation III nuclear power development in China."

AP1000 construction and equipment contracts

Westinghouse and Shaw Group (now CB&I) have an engineering, procurement, commissioning and start-up as well as project supervision contract with SNPTC, the two project companies, and others for the first four reactors (Sanmen & Haiyang), signed in July 2007. Also Shaw has a contract with State Nuclear Power Engineering Corp. Ltd (SNPEC), a SNPTC subsidiary, for technical support for the first two Dafan, Xianning units in Hubei province, including engineering and design management, project controls, quality assurance, construction management and project management.

In April 2007, Westinghouse signed a $350 million contract with Doosan Heavy Industries in Korea for two pressure vessels and four steam generators for Sanmen 1 and Haiyang 1. The pressure vessels for the other two units are being made by Chinese manufacturers: China First Heavy Industries (CFHI, also known as YiZhong) for Sanmen 2 and Shanghai Electric Group Corporation (SEC) for Haiyang 2. Steam generators for Sanmen 2 and Haiyang 2 were manufactured by Harbin Power Equipment Co., Ltd. (HPEC, now Harbin Electric Co, HEC) and SEC, respectively, though a contract for Sanmen 2 steam generators was let to Spain's ENSA in

2011. The Sanmen 1 reactor pressure vessel arrived on site from Doosan in July 2011.

All four steam turbine generators were supplied by Mitsubishi Heavy Industries (MHI). In a $521 million deal, Sanmen Nuclear Power ordered two turbine generator packages from MHI at the end of September 2007, with Shandong Nuclear Power ordering another two early in 2008 for Haiyang. MHI's Takasago Machinery Works manufactured the turbines, including rotors and blades. Mitsubishi Electric Corporation is supplying the generators and Harbin Electric Co (HEC), partnering with MHI, was responsible for turbine casings, piping and associated facilities. The turbines boost the capacity of the reactors from their original 1175 MWe to 1250 MWe gross.

In 2014 Westinghouse said that the second plant at each site saw a 30% reduction in manpower requirements compared with the first unit. The company has also been working on the next eight units anticipated in China and expects about a 50% productivity increase compared with the first two Chinese AP1000 units.

The AP1000 Technology Transfer agreement to SNPTC covered 34 task packages in 7 categories. In November 2010, further contracts were signed between SNPTC and Westinghouse, including one for Westinghouse to provide SNPTC with technical consulting services in research and development of the CAP1400 nuclear power plant, to be developed by SNPTC with Chinese intellectual property rights. Westinghouse said that having shared design technology with SNERDI, it expected 100% localization by 2015.

In January 2011, a further agreement was signed with SNPTC to deploy further AP1000 units, and to extend the 2008 technology cooperation agreement for another two years. SNPTC said the deal also included measures "to develop cooperation in the field of overseas markets." Another agreement was signed with China Baotou Nuclear Fuel Co "to design, manufacture and install fuel fabrication equipment that will enable China to manufacture fuel" for AP1000 units. The latter $35 million contract involves supply and installation of US equipment at Baotou.

In June 2011 Westinghouse confirmed that as part of the earlier construction agreement it would transfer intellectual property rights to SNPTC. In September 2014 Westinghouse said that it expected to be involved in orders for eight further units, followed by 12 more inland and 12 on the coast. It expected “to have 20 AP1000 units under construction in the next five years in China.” In 2018 the first

domestically-produced canned main coolant pump for the AP1000 was produced, these being formerly supplied by Curtis Wright in the USA.

In 2015 SNPTC merged with CPI and took over all its nuclear business as a sub-group of the new State Power Investment Corporation (SPIC) entity. It then claimed that its supply chain could support construction of 6-8 AP/CAP units each year, with up to five suppliers for each key item.

Taishan 1&2 EPR

In February 2007, EDF entered a cooperation agreement with CGN (then: CGNPC) to build and operate a two-unit EPR power station at Yangjiang in Guangdong province. This deal was not expected to involve the technology transfer which is central to the Westinghouse contracts, since the EPR has multiple redundant safety systems rather than passive safety systems and is seen to be more complex and expensive, hence of less long-term interest to China. However, negotiations with Areva and EDF dragged on and in August 2007 it was announced that the EPR project had been shuffled to Taishan (in Guangdong) so that six CPR-1000 units previously planned for that site could be built at Yangjiang as soon as possible.

At a November 2007 ceremony attended by Chinese president Hu Jintao and French president Nicolas Sarkozy in Beijing's Great Hall of the People, Areva initialled an €8 billion contract with CGN for the two EPRs at Taishan plus supply of fuel to 2026 and other materials and services for them. The whole project, including fuel supply, totals €8 billion, of which the nuclear reactors themselves were reported to be about €3.5 billion. Steam turbine generators costing €300 million are included in the larger sum. The Guangdong Development Commission quoted the total investment in both units as CNY 49.85 billion ($7.3 billion). The joint venture partners will put up CNY 16.45 billion and the balance will be borrowed with guarantee from the Central Bank of France. French export credits for the project are reported as €1.7 billion ($2.4 billion), covering purchase of equipment such as steam generators for unit 1 from French suppliers.

In August 2008, EDF and CGN signed the final agreements for the creation of Guangdong Taishan Nuclear Power Joint Venture Company Limited (TNPC). EDF will hold 30% of TNPC for a period of 50 years (the maximum period permitted for a joint venture in China), CGN 70%, though Guangdong Yudean Group (now

Guangdong Energy Group) took over part of this in 2012 during construction to hold 19%. TNPC was responsible for overseeing the building, then own and operate the plant. EDF paid €2.4 billion over four years for its share. (EDF is project manager and architect for the Flamanville 3 EPR project in France, and this initiative consolidates its change in corporate strategy outside France.)

CGN authorized construction at Taishan in July 2008 and first concrete was poured in October 2009, though the official inauguration ceremony was not until 21 December. The first unit was initially expected to be connected to the grid early in 2014 with commercial operation two months later, and the second was to be completed in 2015. The major components for unit 1 were imported: the pressure vessel from Mitsubishi Heavy Industries (MHI) in Japan and the steam generators from Areva's Creusot Forge in France*, but those for unit 2 were all built in China: the pressure vessel by Dongfang Electric (DEC), the steam generators by DEC (two) and Shanghai Electric (two). The Arabelle steam turbines and 1750 MWe generators were purchased separately from Alstom and Dongfang Electric Co respectively. The first generator was shipped by DEC in August 2013. In March 2012 Areva said unit 1 was 69% complete, with reactor pressure vessel delivered. Taishan 1 achieved first criticality and grid connection in June 2018 – the first EPR in the world to do so. In December 2018 unit 1 entered commercial operation. In June 2019 unit 2 was grid connected, and in September it entered commercial operation.

In July 2021 CGN took Taishan 1 offline after damaged fuel rods led to an increase in noble gases within the reactor's primary circuit. Taishan 2 was taken offline in May 2022 to check for similar issues. Taishan 1 resumed operation in August 2022 and unit 2 resumed in June 2023.

* Over 2015-17 the NNSA was involved with France's ASN in assessing manufacturing defects on the Flamanville 3 EPR, particularly the reactor vessel forging.

Earlier in April 2013 Areva, EDF and CGN signed a tripartite agreement for "deeper industrial and commercial cooperation" in building new nuclear power plants and improving all CGN units, though it appears that not more than two further EPR units would be built in China. CGN has 33.5% of the Hinkley Point C project in the UK, where two EPR units are being built by EDF.

Nuclear growth

The first two nuclear power plants in mainland China were at Daya Bay near Hong Kong and Qinshan, south of Shanghai, with construction starting in the mid-1980s. Localization has risen from 1% at Daya Bay to 64% at Ling Ao Phase II, 75% at Hongyanhe, 80% at Ningde and 85% at Yangjiang.

China's concerted nuclear expansion began with the National Development and Reform Commission's (NDRC's) Tenth Economic Plan for the years 2001-2005, with increased self-reliance. (China's first economic plan was in 1953 and began China's centrally planned industrialization under Mao Zedong.) It incorporated the construction of eight nuclear power plants, though the timeline for contracts was extended, putting the last two projects into the 11th plan. The Eleventh Economic Plan for the years 2006-2010 set even more ambitious goals than the Tenth for new nuclear plant construction, and marked a watershed in China's commitment to third-generation reactors, such as the Sanmen plant in Zhejiang province and Haiyang plant in Shandong province (see section above on Reactor technology) as well as maturing of CPR-1000 technology. The 11th Five-Year Plan (2006-10) also had firmer environmental goals than previously, including reduction of 20% in the amount of energy required per unit of GDP, *i.e.* 4% reduction per year.

In 2007, it was announced that three state-owned corporations had been authorized by the NNSA to own and operate nuclear power plants: CNNC, CGN and China Power Investment Corporation (CPI, now SPIC). Any other public or private companies are to have minority shares (up to 25%) in new projects, which proved a severe constraint on the ambitions of the country's main power utilities (including Huaneng, Huadian, Datang and Guodian), all of which have set up nuclear subsidiaries or become involved in nuclear projects. After eight years' experience as minor partners in construction and operation of nuclear plants they may hold a share of 25% or more. Today Huaneng has 49% in Changjiang phase 1 (51% in phase 2), Datang has 44% in Ningde, Huadian has 39% in Fuqing, and Guodian 5% in Haiyang and 49% in Zhangzhou (both main and SMR projects). The Shidaowan HTR-PM project is classed as R&D, so Huaneng has 47.5% and CNEC 32.5%.

The 12th Five-Year Plan (2011-15) included construction start on phase II of Tianwan, Hongyanhe, Sanmen and Haiyang, as well as phase I of inland sites: Taohuajiang, Xianning, and Pengze (two reactors each except for Taohuajiang, with four). By the end of the 12th Five-Year Plan some 25 GWe of new capacity was planned to be operational, making some 40 GWe, and 45 GWe more might be added by the end of the 13th Five Year Plan. The 12th five-year

science & technology plan released in July 2011 called for building a demonstration CAP1400 plant for grid connection in 2015 and the demonstration HTR-PM to begin test operation at Shidaowan before 2015. However, construction starts were delayed following the Fukushima accident, and all the inland plants were put on hold.

More than 16 provinces, regions and municipalities announced intentions to build nuclear power plants in the 12th Five Year Plan (2011-15) so that nuclear plants would be operating or under construction in all coastal provinces except Hebei. Provinces put together firm proposals by 2008 and submitted them to the central government's National Development and Reform Commission (NDRC) for approval during 2009. NRDC consideration is via the new National Energy Administration (NEA). A great many proposals were received, many of which were deferred to the 13th Plan.

The approvals process in China has several stages:

1. Pre-feasibility – a site is selected by the operator and/or the province. Site is qualified for receiving a nuclear plant.
2. 'Road Ticket' – approval by the NRDC and the NEA for preliminary site works, but no firm decision on reactor model or building schedule.
3. NNSA notice/comment – this is scrutinized by the NRDC and the NEA, and the State Council approves a 'Project Application Report'. Excavation works can start.
4. The NNSA investigates further environmental and safety reports and approves first structural concrete. Construction formally starts.
5. After NNSA inspection, the site receives fuel loading authorization.
6. After all hot testing phases, the NNSA issues the business/operating licence.
7. First reactor criticality.

In 2014 CPI had plans to achieve 14 GWe of operational capacity by 2020, with 10 GWe under construction then, at nine sites with 40 units. All of this except phase 1 of Hongyanhe (4xCPR-1000) would be CAP1000.

The complex ownership structure of Chinese nuclear plants is described in Appendix 1: Government Structure and Ownership, and China's considerable heavy engineering and manufacturing capacity is detailed in the information page on Heavy Manufacturing of Power Plants. The capacity at the end of 2013 was for eight sets of reactor equipment per year – over 8000 MWe.

On the people and skills front, 47 colleges and research institutes have nuclear major, and more than 2000 graduate each year (4000 in 10th Five-Year Plan, 11,000 in 11th Five-Year Plan). However, a draft regulation provides that any majority shareholder in a nuclear power project is required to make available 300 qualified people,

with 50% having worked for five years at a nuclear plant. This limits national and provincial electricity investors relative to the three major nuclear utilities.

First and second-wave nuclear plants: varied beginnings to 2010

Daya Bay, Ling Ao Phase I

These are essentially on the one site in Guangdong province, close to Hong Kong. The Daya Bay reactors are standard three-loop French PWR units supplied by Framatome and designated M310, with GEC-Alstom turbines. Electricite de France (EDF) managed construction, starting August 1987, with the participation of Chinese engineers. Commercial operation of the two Daya Bay units was in February and May 1994. There were long outages in 1994-96 when Framatome had to replace major components. Reactor vessel heads were replaced in 2004. The plant produces about 13 TWh per year, with 70% transmitted to Hong Kong and 30% to Guangdong.

Daya Bay (CGN)

The Ling Ao Phase I reactors are virtually replicas of adjacent Daya Bay units. Construction started in May 1997 and Ling Ao 1 started up in February 2002 entering commercial operation in May. Ling Ao 2 was connected to the grid about September 2002 and entered

commercial operation in January 2003. The two Ling Ao reactors use French M310 technology supplied by Framatome (now Areva), but with 30% localization. They are reported to have cost $1800 per kilowatt.

Daya Bay and Ling Ao I & II together comprise the 'Daya Bay nuclear power base' managed by Daya Bay Nuclear Power Operations & Management Co (DNMC), part of China General Nuclear Power Group (CGN). For Ling Ao Phase II, see below.

Qinshan

CNNC's Qinshan 1 in Zhejiang province 100 km southwest of Shanghai, is China's first indigenously-designed and constructed nuclear power plant (though with the pressure vessel supplied by Mitsubishi, Japan). Design of the 300 MWe PWR was by the Shanghai Nuclear Engineering Research & Design Institute (SNERDI). Construction work spanned 6.5 years from March 1985, with first grid connection in December 1991. It was shut down for 14 months for major repairs from mid-1998. CNNP applied for a 20-year licence extension in 2016, to take the unit past 2021, and this was approved in 2021. It also invited the IAEA to conduct a Safety Aspects for Long-Term Operation (SALTO) peer review of the reactor, which was done in 2017. The IAEA reviewed the company's response to the SALTO recommendations and suggestions in May 2019, reporting to the company and NNSA.

In October 2007, Qinshan 1 was shut down for a major upgrade. The entire instrumentation and control system was replaced, along with the reactor pressure vessel head and control rod drives. Areva NP supervised the work, with a view to extending the operating lifetime beyond the original 30 years, and in January 2020 the NNSA concluded a safety review which said it was safe to uprate and extend the operating lifetime by 20 years, to 2041. In April 2019, an upgrade to the turbine increased capacity, and in March 2020 the NEA approved a licence amendment to increase nominal capacity from 310 to 330 MWe.

Qinshan Phase II units 1&2 are locally-designed and constructed two-loop PWR reactors, scaled up from Qinshan 1, and designated CNP-600. Local content was 55%. Unit 1 started up at the end of 2001 and entered commercial operation in April 2002. Unit 2 started up in March 2004, with commercial operation in May 2004. In March 2025 Qinshan II-2 returned to service after a refuelling and maintenance outage during which the unit was uprated by 33 MWe.

Units 3&4 are similar, with local content of 77%. Construction of the second stage of Qinshan Phase II was formally inaugurated at the end of April 2006, though first concrete had been poured for unit 3 in March. That for unit 4 was poured in January 2007. After 53 months' construction, unit 3 was grid-connected on 1 August 2010, and entered commercial operation 12 weeks later[6]. Unit 4 was grid-connected in November 2011 and entered commercial operation in April 2012. CNNC claims that Qinshan phase II "is the first independently-designed, built, managed and operated large commercial nuclear power station in China."

In 2004, CNNC announced that the next two Qinshan units would be 1000 MWe indigenous units (in effect Fangjiashan, adjacent to Qinshan 1, has taken over this role).

Qinshan Phase III units 1&2 use the Candu 6 pressurized heavy water reactor (PHWR) technology, with Atomic Energy of Canada (AECL) being the main contractor of the project on a turnkey basis. Construction began in 1997 and unit 1 started up in September 2002 and unit 2 in April 2003. They are each about 678 MWe net. Nucleoelectrica Argentina was in March 2022 awarded a contract by CNNC for the planned refurbishment of the two units. CNNC plans to extend their operation for another 30 years with work starting in either 2028 or 2029.

In December 2022 China's first industrial heating project from a nuclear power plant was completed at Qinshan. The project guarantees 24-hour heat supply of around 288,000 gigajoules per year, saving about 10,000 tonnes of standard coal and 24,000 tonnes of carbon emissions annually.

In May 2025 China's first batch of the carbon-14 isotope produced at a commercial reactor was dispatched from the Qinshan nuclear power plant in Zhejiang province.

Tianwan phase I

Tianwan Phase I at Lianyungang city in Jiangsu province is a Russian AES-91 power plant (with two 1060 MWe V-428 VVER reactors) constructed under a cooperation agreement between China and Russia – the largest such project ever. The cost is reported to be $3.2 billion, with China contributing $1.8 billion of this. Completion was delayed due to corrosion in the steam generators which resulted in some tubes having to be plugged with a net loss of capacity of about 2%. The first unit was grid connected in May 2006 and put into commercial operation in June 2007. The

second was grid connected in May 2007, with commercial operation in August 2007. Design life is 40 years.

Ling Ao Phase II (Ling Dong)

While the bidding process for the delayed Generation III plants from overseas vendors was in train over more than two years (see section above on Embarking upon Generation III plants), the China General Nuclear Power Group (CGN) signed contracts with Chinese designers and manufacturers for two reactors as Phase II of the Ling Ao power station (also known as Ling Dong). Construction started in December 2005 with the 1080 MWe (gross), 1037 MWe (net) units. They are transitional M310 - CPR-1000 units of 1037 MWe net. Unit 1 is about 50% localized and unit 2 is 70% localized, built under the project management of China Nuclear Power Engineering Corporation (CNPEC), part of CGNPC. Low-speed Arabelle turbine-generator sets were provided by Alstom or DEC. In June 2009, the first Chinese-made reactor pressure vessel for a 1000 MWe class reactor was delivered for unit 2, from Dongfang (Guangzhou) Heavy Machinery Co. Unit 1 started up in June 2010 with grid connection in mid-July, 54 months after construction start, and entered commercial operation in September. Unit 2 was grid connected in May, 60 months after construction start, and commenced commercial operation on schedule in August 2011. The plant is managed by Daya Bay Nuclear Power Operations & Management Co (DNMC).

Operable nuclear power reactors

Reactor name	Province	Project control	Model	Reactor Type	Net capacity (MWe)	Construction start
Changjiang 1	Hainan	Huaneng & CNNC	CNP-600	PWR	601	2010-04-25
Changjiang 2	Hainan	Huaneng & CNNC	CNP-600	PWR	601	2010-11-21
Daya Bay 1	Guangdong	CGN	M310	PWR	944	1987-08-07
Daya Bay 2	Guangdong	CGN	M310	PWR	986	1988-04-07
Fangchenggang 1	Guangxi	CGN	CPR-1000	PWR	1000	2010-07-30

Reactor name	Province	Project control	Model	Reactor Type	Net capacity (MWe)	Construction start
Fangchenggang 2	Guangxi	CGN	CPR-1000	PWR	1000	2010-12-23
Fangchenggang 3	Guangxi	CGN	HPR1000	PWR	1105	2015-12-24
Fangchenggang 4	Guangxi	CGN	HPR1000	PWR	1105	2016-12-23
Fangjiashan 1	Zhejiang	CNNC	CPR-1000	PWR	1012	2008-12-26
Fangjiashan 2	Zhejiang	CNNC	CPR-1000	PWR	1012	2009-07-17
Fuqing 1	Fujian	CNNC & Huadian	CPR-1000	PWR	1000	2008-11-21
Fuqing 2	Fujian	CNNC & Huadian	CPR-1000	PWR	1000	2009-06-17
Fuqing 3	Fujian	CNNC & Huadian	CPR-1000	PWR	1000	2010-12-31
Fuqing 4	Fujian	CNNC & Huadian	CPR-1000	PWR	1000	2012-11-17
Fuqing 5	Fujian	CNNC & Huadian	Hualong One	PWR	1075	2015-05-07
Fuqing 6	Fujian	CNNC & Huadian	HPR1000	PWR	1075	2015-12-22
Haiyang 1	Shandong	SPIC	AP1000	PWR	1170	2009-09-24
Haiyang 2	Shandong	SPIC	AP1000	PWR	1170	2010-06-20
Hongyanhe 1	Liaoning	CGN & SPIC	CPR-1000	PWR	1061	2007-08-18
Hongyanhe 2	Liaoning	CGN & SPIC	CPR-1000	PWR	1061	2008-03-28
Hongyanhe 3	Liaoning	CGN & SPIC	CPR-1000	PWR	1061	2009-03-07
Hongyanhe 4	Liaoning	CGN & SPIC	CPR-1000	PWR	1061	2009-08-15
Hongyanhe 5	Liaoning	CGN & SPIC	ACPR-1000	PWR	1061	2015-03-29

Reactor name	Province	Project control	Model	Reactor Type	Net capacity (MWe)	Construction start
Hongyanhe 6	Liaoning	CGN & SPIC	ACPR-1000	PWR	1061	2015-07-24
Ling Ao 1	Guangdong	SPIC	M310	PWR	950	1997-05-15
Ling Ao 2	Guangdong	SPIC	M310	PWR	950	1997-11-28
Ling Ao 3	Guangdong	SPIC	CPR-1000	PWR	1007	2005-12-15
Ling Ao 4	Guangdong	SPIC	CPR-1000	PWR	1007	2006-06-15
Ningde 1	Fujian	CGN & Datang	CPR-1000	PWR	1018	2008-02-18
Ningde 2	Fujian	CGN & Datang	CPR-1000	PWR	1018	2008-11-12
Ningde 3	Fujian	CGN & Datang	CPR-1000	PWR	1018	2010-01-08
Ningde 4	Fujian	CGN & Datang	CPR-1000	PWR	1018	2010-09-29
Qinshan 1	Zhejiang	CNNC	CNP-300	PWR	326	1985-03-20
Qinshan 2 1	Zhejiang	CNNC	CNP-600	PWR	623	1996-06-02
Qinshan 2 2	Zhejiang	CNNC	CNP-600	PWR	623	1997-04-01
Qinshan 2 3	Zhejiang	CNNC	CNP-600	PWR	623	2006-04-28
Qinshan 2 4	Zhejiang	CNNC	CNP-600	PWR	623	2007-01-28
Qinshan 3 1	Zhejiang	CNNC	CANDU 6	PHWR	677	1998-06-08
Qinshan 3 2	Zhejiang	CNNC	CANDU 6	PHWR	677	1998-09-25
San'ao 1	Zhejiang	CGN	Hualong One	PWR	1117	2020-12-31
Sanmen 1	Zhejiang	CNNC	AP1000	PWR	1157	2009-04-19

Reactor name	Province	Project control	Model	Reactor Type	Net capacity (MWe)	Construction start
Sanmen 2	Zhejiang	CNNC	AP1000	PWR	1157	2009-12-15
Shidaowan Guohe One 1	Shandong	SPIC & Huaneng	CAP1400	PWR	1400	2019-06-19
Shidaowan HTR-PM 1	Shandong	Huaneng	HTR-PM	HTGR	150	2012-12-09
Taipingling 1	Guangdong	CGN	Hualong One	PWR	1116	2019-12-26
Taishan 1	Guangdong	CGN	EPR-1750	PWR	1660	2009-11-18
Taishan 2	Guangdong	CGN	EPR-1750	PWR	1660	2010-04-15
Tianwan 1	Jiangsu	CNNC	VVER V-428	PWR	1000	1999-10-20
Tianwan 2	Jiangsu	CNNC	VVER V-428	PWR	1000	2000-09-20
Tianwan 3	Jiangsu	CNNC	VVER V-428M	PWR	1060	2012-12-27
Tianwan 4	Jiangsu	CNNC	VVER V-428M	PWR	1060	2013-09-27
Tianwan 5	Jiangsu	CNNC	ACPR-1000	PWR	1060	2015-12-27
Tianwan 6	Jiangsu	CNNC	ACPR-1000	PWR	1060	2016-09-07
Yangjiang 1	Guangdong	CGN	CPR-1000	PWR	1000	2008-12-16
Yangjiang 2	Guangdong	CGN	CPR-1000	PWR	1000	2009-06-04
Yangjiang 3	Guangdong	CGN	CPR-1000	PWR	1000	2010-11-15
Yangjiang 4	Guangdong	CGN	CPR-1000	PWR	1000	2012-11-17
Yangjiang 5	Guangdong	CGN	ACPR-1000	PWR	1000	2013-09-18
Yangjiang 6	Guangdong	CGN	ACPR-1000	PWR	1000	2013-12-23

Reactor name	Province	Project control	Model	Reactor Type	Net capacity (MWe)	Construction start
Zhangzhou 1	Fujian	CNNC	Hualong One	PWR	1126	2019-10-16
Zhangzhou 2	Fujian	CNNC	Hualong One	PWR	1126	2020-09-04

In addition, the China Experimental Fast Reactor (CEFR) is grid-connected and occasionally producing 20 MWe net. It is included in IAEA figures for operational reactors.

Further nuclear plants: operating, under construction and planned

Between 2019 and 2023, approval was given for the construction of 35 new reactors in China.

In August 2024 it approved a further five nuclear power projects – Xuwei Phase I, Lufeng Phase I, Zhaoyuan Phase I, San'ao Phase II and Bailong Phase I – with a total of 11 reactors.

In April 2025 China's State Council approved five nuclear power projects – Fangchenggang Phase III, Haiyang Phase III, Sanmen Phase III, Taishan Phase II and Xiapu Phase I – with a total of 10 reactors, including eight Hualong One units.

Nuclear reactors under construction

Reactor name	Province	Project control	Model	Reactor type	Net capacity (MWe)	Construction start
Bailong 1	Guangxi	SPIC	CAP1000	PWR	1161	2025-12-25
Changjiang SMR-1	Hainan	CNNC	ACP100	PWR	125	2021-07-13
Changjiang-3	Hainan	Huaneng & CNNC	Hualong One	PWR	1100	2021-03-31
Changjiang-4	Hainan	Huaneng & CNNC	Hualong One	PWR	1100	2021-12-28
Haiyang 3	Shandong	SPIC	CAP1000	PWR	1161	2022-07-07
Haiyang 4	Shandong	SPIC	CAP1000	PWR	1161	2023-04-22

Reactor name	Province	Project control	Model	Reactor type	Net capacity (MWe)	Construction start
Jinqimen 1	Zhejiang	CNNC	Hualong One	PWR	1120	2025-08-10
Lianjiang 1	Guangdong	SPIC	CAP1000	PWR	1161	2023-09-27
Lianjiang 2	Guangdong	SPIC	CAP1000	PWR	1161	2024-04-26
Lufeng 1	Guangdong	CGN	CAP1000	PWR	1161	2025-02-24
Lufeng 2	Guangdong	CGN	CAP1000	PWR	1161	2025-12-22
Lufeng 5	Guangdong	CGN	Hualong One	PWR	1116	2022-09-08
Lufeng 6	Guangdong	CGN	Hualong One	PWR	1116	2023-08-26
Ningde 5	Fujian	CGN & Datang	Hualong One	PWR	1116	2024-07-28
Ningde 6	Fujian	CGN & Datang	Hualong One	PWR	1116	2025-12-16
San'ao 2	Zhejiang	CGN	Hualong One	PWR	1117	2021-12-30
San'ao 3	Zhejiang	CGN	Hualong One	PWR	1117	2025-11-19
Sanmen 3	Zhejiang	CNNC	CAP1000	PWR	1163	2022-06-28
Sanmen 4	Zhejiang	CNNC	CAP1000	PWR	1163	2023-03-22
Shidaowan 1	Shandong	SPIC & Huaneng	Hualong One	PWR	1134	2024-07-28
Shidaowan 2	Shandong	SPIC & Huaneng	Hualong One	PWR	1134	2025-05-08
Shidaowan Guohe One 2	Shandong	Huaneng & CNNC	CAP1400	PWR	1400	2020-04-21
Taipingling 2	Guangdong	CGN	Hualong One	PWR	1116	2020-10-15
Taipingling 3	Guangdong	CGN	Hualong One	PWR	1116	2025-06-10
Tianwan 7	Jiangsu	CNNC	VVER V-491	PWR	1100	2021-05-19
Tianwan 8	Jiangsu	CNNC	VVER V-491	PWR	1100	2022-02-25

Reactor name	Province	Project control	Model	Reactor type	Net capacity (MWe)	Construction start
Xiapu 1	Fujian	CNNC	CFR-600	FBR	600	2017-12-29
Xiapu 2	Fujian	CNNC	CFR-600	FBR	600	2020-12-27
Xin'an HTR-PM 600-1	Shandong	CNNC	HTR-PM600	HTGR	600	2024-06-10
Xin'an HTR-PM 600-2	Shandong	CNNC	HTR-PM600	HTGR	600	2024-12-09
Xudabao 1	Lianoning	CNNC & Datang	CAP1000	PWR	1000	2023-11-03
Xudabao 2	Lianoning	CNNC & Datang	CAP1000	PWR	1000	2024-07-17
Xudabao 3	Lianoning	CNNC & Datang	VVER V-491	PWR	1200	2021-07-28
Xudabao 4	Lianoning	CNNC & Datang	VVER V-491	PWR	1200	2022-05-19
Xuwei 1	Jiangsu	CNNC	Hualong One	PWR	1116	2026-01-16
Zhangzhou 3	Fujian	CNNC	Hualong One	PWR	1126	2024-02-22
Zhangzhou 4	Fujian	CNNC	Hualong One	PWR	1129	2024-09-27
Zhaoyuan 1	Shandong	CGN	Hualong One	PWR	1116	2025-11-18

* Dates from satellite imagery showing the unit under construction. Official first concrete is yet to be confirmed and may be significantly earlier.

Nuclear reactors planned

Units	Province	MWe gross	Reactor model	Project control	State approval
Bohai Shipyard FNPP	Liaoning	1x60	ACPR50S	CGN	
Lufeng 3-4	Guangdong	2x1250	CAP1000	CGN	August 2024
Huaneng Xiapu 3-6	Fujian	4x1200	Hualong One	Huaneng & CNNC	April 2025

Units	Province	MWe gross	Reactor model	Project control	State approval
Haixing 1&2	Hebei	2x1250	CAP1000	CNNC & Huadian	
Fangchenggang 5&6	Guangxi	2x1208	Hualong One	CGN	April 2025
Bailong 2	Guangxi	1x1250	CAP1000	SPIC	August 2024
Changjiang SMR 2	Hainan	1x125	ACP100	CNNC	
Taishan 3&4	Guangdong	2x1200	Hualong One	CGN	April 2025
Xianning 1&2	Hubei (inland)	2x1250	CAP1000	CGN	
Zhaoyuan 2	Shandong	1x1200	Hualong One	CGN	August 2024
Zhuanghe 1&2	Liaoning	2x1200	Hualong One	Datang	
Zhuanghe 3-6	Liaoning	4x1200	Hualong One	Datang	
Cangnan/San'ao 4-6	Zhejiang	3x1200	Hualong One	CGN/Huaneng	
Sanmen 5&6	Zhejiang	2x1215	Hualong One	CNNC	
Shidaowan 3&4	Shandong	2x1250	Hualong One	China Huaneng & CNNC	
Haiyang 5&6	Shandong	2x1250	CAP1000	CPI	April 2025
Huizhou/Taipingling 4	Guangdong	1x1200	Hualong One	CGN	January 2024
Xiangshan Jinqimen 2	Zhejiang	1x1150	Hualong One	CNNC	January 2024
Xiapu HTR	Fujian	1x600	HTR-PM600	Huaneng	
Xuwei 2	Jiangsu	1x1200	Hualong One	CNNC	August 2024
Xuwei HTR 1	Jiangsu	1x600	HTR-PM600	CNNC	
Xin'an 1&2	Shandong	2x1200	Hualong One	CNNC	

Units	Province	MWe gross	Reactor model	Project control	State approval
Laiyang 1&2	Shandong	2x1500	CAP1400	SPIC	
Total: 42		48,131			

It is likely that most planned CAP1000 units will be displaced by Hualong One.

Nuclear power units proposed

Units	Province	MWe gross	Expected model	Project control
Nanchong (Nanchun, Sanba) 1-4	Sichuan	4x1200	Hualong One	CGN
Pengze 1-4	Jiangxi	4x1250	CAP1000	CPI
Taohuajiang 1-4	Hunan (inland)	4x1250	CAP1000	CNNC
Bailong 3-6	Guangxi	4x1400	CAP1400	CPI
Pingnan/Baisha 1&2	Guangxi	2x1250	CAP1000	CPI
Pingnan/Baisha 3&4	Guangxi	2x1250	CAP1000	CPI
Xudabao 5&6	Liaoning	2x1250	CAP1000	CNNC with Datang
Yingtan 1-4	Jiangxi	4x1250	CAP1000	Huaneng
Nanyang 1-6	Henan	6x1250	CAP1000 (if CPI)	CNNC (or CPI)
Changde (Chenzhou, Hengyang)	Hunan	4x1200	Hualong One	CNNC & Guodian?, CGN
Zhangzhou 5&6	Fujian	2x1200	Hualong One	CNNC & Guodian
Jiyang/Chizhou 1&2	Anhui	2x1250	CAP1000	CNNC
Fuling 1&2	Chongqing	2x1250	CAP1000	CPI
Jingyu 1&2	Jilin	2x1250	CAP1000	CPI & Guodian
Donggang 1&2	Liaoning	2x1200	Hualong One?	Huadian
Wuhu 1&2	Anhui	2x1250	CAP1000	CGN

Units	Province	MWe gross	Expected model	Project control
Ningdu 1&2	Jiangxi	2x125	ACP100	CNNC & Guodian
Yanjiashan/Wanan/Ji'an 1&2	Jiangxi	2x1250	CAP1000	CNNC
Shaoguan 1-4	Guangdong (inland)	4x1250	CAP1000	CGN
Huizhou/Taipingling units 5&6	Guangdong	2x1200	Hualong One	CGN
Zhaoyuan 3-6	Shandong	4x1200	Hualong One	CGN
Haixing 3-6	Hebei	4x1250	CAP1000	CNNC & Huadian
Jiyang/Chizhou 3&4	Anhui	2x1250?	CAP1000?	CNNC
Longyou/Zhexi 1-4	Zhejiang	4x1250	CAP1000	CNNC
Haijia/Haifeng 1&2	Guangdong	2x1250	CAP1000	CGN
Fuling 3&4	Chongqing	2x1250	CAP1000	CPI
Jingyu 3&4	Jilin	2x1250	CAP1000	CPI & Guodian
Songjiang 1&2	Shanghai	2x1250?	CAP1000	CGN & Guodian
Wuhu 3&4	Anhui	2x1250	CAP1000	CGN
Heyuan/Jieyang 1-4	Guangdong	4x1250	CAP1000	CNNC? CPI?
Haiyang 7&8	Shandong	2x1250	CAP1000	CPI
Hengren 1-4	Liaoning	4x1250	CAP1000	CPI
Lianjiang 3-6	Guangdong	4x1400	CAP1400	SPIC
Xiangtan	Hunan	4x1250	CAP1000	Huadian?
Donggang 3&4	Liaoning	2x1200	Hulaong One	Huadian?
Pulandian 1-4	Liaoning	4x1250	CAP1000	Huaneng
Shizu	Chongqing	2x		CNNC
Qiaofushan	Hebei	2x		CNNC
Guangshui	Hubei	4x1250	CAP1000	CGN
Zhingxiang	Hubei	4x1250	CAP1000	CNNC, Datang

Units	Province	MWe gross	Expected model	Project control
Hebaodao	Guangdong	2x		CNNC?
Yibin	Sichuan	2x1250	CAP1000	CNNC
Gulei 1&2	Fujian	2x125	ACP100	CNNC-CNEC
Hengfeng 1&2	Jiangxi	2x125	ACP100	CNNC & Guodian
Tongren	Guizhou	2x1250	CAP1000	CGN
Jiamusi	Heilongjiang	2x1200	Hualong One	Huaneng & CNNC, or CGN
Xuwei 3&4	Jiangsu	2x1200	Hualong One	CNNC
Xuwei HTR 2	Jiangsu	1x600	HTR-PM600	CNNC
Xin'an 3&4	Shandong	2x1200	Hualong One	CNNC
Xin'an HTR 1&2	Shandong	2x600	HTR-PM600	CNNC
Laiyang 3-6	Shandong	4x1500	CAP1400	SPIC
Xiangshan Jinqimen 3-6	Zhejiang	4x1200	Hualong One	CNNC
Jiaodong Shipyard FNPP	Shandong	1x125	ACP100S	CNNC
Total: 146 units				

It is likely that many proposed CAP1000 units listed in this Table will be replaced by Hualong One or Guohe One (CAP1400). Some of these entries are based on uncertain information.

Hongyanhe, LHNP

This is the first nuclear power station receiving central government approval to build four units at the same time, and the first in northeast China. Construction of the first unit of the Hongyanhe nuclear power plant in Dalian, Liaoning, started in August 2007. It is the first nuclear power project in the 11th Five-Year Plan, with owner and operator being Liaoning Hongyanhe Nuclear Power Co, a joint venture of CGN and CPI (now SPI) – 45% each – with Dalian

Construction Investment Group. It uses CPR-1000 reactors for units 1-4, and in the cold water there they are each 1118.79 MWe gross.

The National Nuclear Safety Administration (NNSA) issued a construction licence for units 3&4 in March 2009, and first concrete for unit 3 was poured soon afterwards. CGN will be responsible for the project construction and the operation of the first five years after commercial operation, with full participation of CPI. The cost of all four 1080 MWe CPR-1000 units in the first construction phase is put at CNY 50 billion (US$ 6.6 billion). China Nuclear Power Engineering Corporation (CNPEC), part of CGN, is managing the project. Shanghai Electric won a $260 million contract for equipment and Alstom providing the four low-speed Arabelle turbine-generator sets for $184 million. Localization is above 70% for units 1&2 and over 80% for units 3&4.

First power from unit 1 was expected in July 2012, but after delays over 2011 it started up in January 2013 and was grid connected in February, with commercial operation in June. Unit 2 started up in October 2013, was grid connected in November, with commercial operation in May 2014. Unit 3 started up in October 2014 and was grid connected in March 2015 (72 months construction) and entered commercial operation in August. Unit 4 started up in March 2016 and was grid connected early April (78 months construction, after fuel loading was suspended for several months) and due to low demand it was not due to start full commercial operation until 2017, but this was brought forward to September 2016. The project incorporates a 10,080 m^3/day seawater desalination plant using waste heat to provide cooling water.

In September 2016 CGN announced completion of phase one of Hongyanhe, “Northeast China’s first nuclear plant and largest energy project.”

In May 2010, the NRDC approved preliminary work on the CNY 25 billion two-unit second phase of the plant (units 5&6), and site work began in July. The National Nuclear Safety Administration (NNSA) and the Environment Ministry approved the project in September 2010, construction start was expected 2011 but following a State Council announcement and final NNSA construction licence, that for unit 5 was in March 2015 and for unit 6, July 2015. NDRC approval was reported in September 2014 and again in February 2015, using ACPR1000 reactors. Localization was expected to be above 80%. In June 2021 unit 5 was connected to the grid and it began commercial operation in August 2021. In May 2022 unit 6 was connected to the grid and it began commercial operation in June

2022. Unit 6 will provide district heating covering a total area of 242,000 m^2 to the nearby city of Hongyanhe during the winter.

Ningde, NDNP

Construction of CGN's six-unit Ningde nuclear power plant commenced in 2008. This is on three islands in Fuding city in northeast of Fujian province, and the first construction phase comprised four CPR-1000 units. Ningde Nuclear Power Co Ltd (NDNP) was set up in 2006 as a joint investment of CGN (46%), China Datang Corporation (44%) and Fujian Energy Group Co., Ltd. The project was approved by the National Development & Reform Commission (NDRC) in September 2006, and local content was about 75% for units 1&2 and 85% for units 3&4. It marked a significant step into nuclear power for Datang.

Construction of the first unit started in February 2008, and it was grid connected in December 2012 after 58 months. It was declared in commercial operation in April 2013. First concrete for the second unit was in November 2008, it was grid connected in January 2014, with commercial operation in May. Construction start for the third was in January 2010 and for the fourth at the end of September 2010. Grid connection for unit 3 was in March 2015, with commercial operation in June. Grid connection for unit 4 was in March 2016 with commercial operation in July. Total cost for four units was put at CNY 52 billion ($7.6 billion). Dongfang Electric supplied turbine generators for units 1-4, using Alstom Arabelle low-speed technology, at least for units 3&4. The pressure vessel and steam generators for unit 1 are from Dongfang (DFHM), those for unit 2 are from Shanghai Electric (SEC), those for units 3&4 from China First (CFHI).

In February 2014 the NEA approved preliminary work for units 5&6, which are set apart from phase 1 units 1-4. In January 2019 CGN approved an EPC agreement for units 5&6. Ownership of these is CGN (43%), China Datang Corporation (47%) and Fujian Energy Group (10%). The State Council approved construction of the two units in July 2023 and first concrete was poured for unit 5 in July 2024 and for unit 6 in December 2025.

Fuqing

Construction of the six-unit Fuqing nuclear power plant 170 km south of Ningde also commenced in 2008 at Qianxe, Fuqing city in

Fujian, between Fuzhou and Putian. The Fujian Fuqing Nuclear Co Ltd was set up in May 2006 by CNNC with 39% held by China Huadian Corp and 10% by Fujian Investment & Development Co Ltd. The CNNC share is held through China National Nuclear Power Co Ltd (CNNP), which used CGN's CPR-1000 reactors for units 1-4 since alternatives were not licensed. CNNC calls these M310+.

Construction of the project was by China Nuclear Power Engineering Co. (CNPE) and the reactor pressure vessels were supplied by China First Heavy Industries, as for Fangjiashan. In June 2008, Dongfang Electric Group announced a CNY 5 billion ($725 million) contract for Alstom Arabelle low-speed steam turbine generators for the Fuqing and Fangjiashan plants. Units 3&4 have Areva instrumentation and control systems.

Late in 2010, CNNC was proposing the CNP1000 for units 5&6, noting "pre-project under way". In October 2011 CNNC said that units 5&6 would be the first ACP1000 units, and in December 2013 it was announced that they would have Areva-Siemens instrumentation and control systems and DEC steam generators. In January 2014 CNNC said that they would be CNNC's first Hualong One units.

First concrete for unit 1 was poured in November 2008, for unit 2 in June 2009, for unit 3 in December 2010, and for unit 4 in September or October 2012, almost immediately after NNSA authorization. Total expected cost for all six was CNY 88 billion ($14.3 billion). Unit 1 started up in July 2014, was grid connected in August after 69 months' construction and entered commercial operation in November. Unit 2 started up in July and was grid connected in August 2015 after 74 months' construction and started commercial operation in October. Unit 3 started up in July 2016, was grid-connected in September after 70 months' construction and achieved commercial operation in October. Unit 4 started up in July 2017, was grid connected at the end of that month, and achieved commercial operation in September.

Construction of unit 5, China's first Hualong One, started in May 2015, and unit 6 construction started in December. Fuel loading of unit 5 began in September 2020, it was connected to the grid in November (66 months after first concrete), and commercial operation was in January 2021. Unit 6 was connected to the grid in January 2022 and began commercial operation in March.

Fangchenggang

The Fangchenggang nuclear power project is located within the Beibu Gulf (Beibu Wan) Economic Zone on the west coast of Qinzhou Bay, about 10 km east of the coastal city of Fangchenggang in the Guangxi Autonomous Region (60 km from the Vietnam border in south China). It is sometimes referred to as being in 'Western China'. Following an agreement in July 2006, the first stage (two 1080 MWe CPR-1000 units out of six planned) of the plant was approved by NDRC in October 2008, and again in July 2010. First concrete for unit 1 was poured in July 2010, and for unit 2, late in 2011. About 87% of the first two units was sourced in China.

In October 2009, a general construction contract was signed with CNPEC for units 1&2. Guangxi Fangchenggang Nuclear Power Co., Ltd., a joint venture between China Guangdong Nuclear Power Group (61%) and Guangxi Investment Group (39%), is responsible for the construction and operation. CPI was earlier involved. The first unit achieved criticality and was grid-connected in October 2015 and began commercial operation in January 2016, the second was grid-connected in July 2016. A cost of CNY 26 billion ($3.87 billion) was envisaged for these two. In 2011 it was reported that Guangxi wanted to allow for exports to Vietnam. (There is also a Fangchenggang supercritical 2400 MWe coal-fired power station operated by CLP Guangxi Fangchenggang Power Company Limited, a 70:30 equity-basis joint venture between China Light & Power and Guangxi Water & Power Engineering (Group) Co., Ltd.)

CGN from September 2012 planned that stage 2, units 3&4, should be the initial ACPR1000 demonstration units, with construction start in 2014. However, no authorization was forthcoming, and since January 2014 these are to be CGN's inaugural Hualong HPR1000 units. Construction of unit 3 started in December 2015 and unit 4 in December 2016. These two were to be the reference plant for CGN's plans to build the Bradwell B nuclear power plant in the UK. In May 2015 CGN ordered the steam generators from DEC, which subcontracted to BWX Technologies in Canada among others. The cost is put at about CNY 30 billion ($4.9 billion). Fuel loading for unit 3 began in December 2022 and it was connected to the grid in January 2023. Unit 4 was connected to the grid in April 2024.

Phase II units 5&6 were planned as AP1000, but are now to be Hualong One/HPR1000. Construction of units 5&6 were approved by the State Council in April 2025.

Earlier in December 2015 CGN and Guangxi Investment Group signed an agreement with Ratch China Power Ltd, a subsidiary of

Thailand's Ratchaburi Electricity Generating Holding Public Company, a subsidiary of the state-owned Electricity Generating Authority of Thailand (EGAT), to establish a joint venture – Guangxi Fangchenggang Nuclear Power (II) Co Ltd – to develop, construct and operate Phase II of the Fangchenggang plant. Ratch China will hold 10%, CGN 51% and Guangxi 39%. Thailand has added Hualong One to its technology shortlist. Ratch will invest THB 7.5 billion ($208 million) in the project over the five years to 2021 under a 30-year power purchase contract.

Yangjiang, YNPS

Yangjiang city in western Guangdong province had originally been earmarked for the country's first Generation III plants (see section above on Embarking upon Generation III plants). After plans changed in the light of pressing generation needs in the region, Yangjiang will be the second nuclear power base of the China General Nuclear Power Group (CGN). Development of all six units of the Yangjiang plant was approved in 2004, with CPR-1000 later confirmed as the basic technology for it. Local content is about 83% for units 1&2, and is expected to be 85% for units 3&4 and maybe 90% for 5&6 of the evolved design – the first ACPR1000 reactors. Total cost was expected to be CNY 73 billion ($11.5 billion).

Construction of the first of two CPR-1000 units by CNPEC started in December 2008. Unit 1 criticality was achieved in December 2013, with grid connection at the end of the month and commercial operation in March 2014. Unit 2 was grid connected in March 2015 after 67 months' construction and entered commercial operation in June.

Construction of unit 3 started in November 2010, unit 4 in November 2012, immediately after NNSA authorization, then the final two (as the second construction phase) followed from September 2013. Units 3&4 are the first of an improved CPR-1000 design sometimes referred to as CPR-1000+. Unit 3 criticality was in October 2015, with grid connection later that month, so 59 months' construction time. Commercial operation was from January 2016. Unit 4 grid connection was in January 2017, after 50 months' construction time, and commercial operation was in March. Construction of unit 5 as an ACPR1000 began in mid-September 2013, with grid connection in May 2018 after construction of 56 months. Unit 6 construction start was late in December 2013.

Yangjiang 1-6 will be operated under Yangjiang Nuclear Power Co Ltd (YJNPC) management. In July 2010, Hong Kong-based power utility China Light and Power (CLP) agreed to take a 17% stake in Yangjiang – the equivalent of one reactor. However, in September 2013 CGN halted negotiations with CLP, following delays due China's review of nuclear safety. In October 2016 CGN Power called for bids for 17% of YJNPC, at CNY 5 billion, and CLP submitted a successful tender then. The utility Guangdong YuDian Group Co holds another 17%, while CGN Group and small subsidiaries retain 66% equity.

Fangjiashan

Construction of CNNC's Fangjiashan plant started at the end of December 2008. It is close to the Qinshan plant in Zhejiang province near Shanghai and essentially an extension of it, using two CPR-1000 reactors, designated M310+ by CNNC. Construction of the CNY 26 billion ($4.2 billion) project was by China Nuclear Power Engineering Co. (CNPE) and the reactor pressure vessels are supplied by China First Heavy Industries, as for Fuqing. In June 2008, Dongfang Electric Group announced a CNY 5 billion ($725 million) contract for Alstom Arabelle low-speed steam turbine generators for the Fuqing and Fangjiashan plants. Localization is 80%. First criticality of unit 1 was in October 2014 with grid connection in November and commercial operation in December 2014 and was grid connected in mid-January 2015. Unit 2 started up in December 2014, with grid connection in mid-January 2015 and commercial operation in February. The project is 72% owned by CNNC, with the remainder held by Zhejiang Provincial Energy Group Co Ltd. The two units bring the Qinshan total effectively to nine, with 6540 MWe.

Sanmen

At the end of 2006, the Westinghouse AP1000 reactor design was selected for Sanmen in Zhejiang province (and for Yangjiang in Guangdong province, with the latter site changed to Haiyang). Contracts with Westinghouse and Shaw for two units were signed in July 2007. Site works under CNNC commenced in February 2008 and an engineering, procurement and construction (EPC) contract was signed in March 2009 between SNPTC and CNNC and China Nuclear Engineering & Construction Group (CNEC/CNECC) for both units, which is overseen by Westinghouse and Shaw (now CB&I).

Other stakeholders are Zhejiang Provincial Energy Group Co Ltd, CPI Nuclear Power Co Ltd, and China Huadian Corp.

Construction on Sanmen 1 – the world's first AP1000 unit – officially commenced on 19 April 2009. The reactor pressure vessel, from Doosan, was installed in September 2011. The reactor was expected to begin operation in December 2014 with unit 2 less than a year later. Construction on unit 2 commenced in mid-December 2009. The pressure vessel and steam generators were made in China. MHI supplied the turbine generators for both units. CNEA in 2013 said the cost would be about $2610/kW. In March 2014 NNSA said that start-up would be delayed to December 2015 due to design changes and problems with the US-supplied main coolant pumps which created further delay. Following successful tests, the modified pumps were shipped from Curtis-Wright to the site in October 2015. Sanmen 1 was grid connected in June 2018 and commenced commercial operation in September 2018, delivering 1157 MWe net. Unit 2 was grid connected in August 2018 and entered commercial operation in November. The original expectation was for 68 months construction, in fact it took 110 months for unit 1 and 104 months for unit 2. See section on Embarking upon Generation III plants above.

In April 2022 the State Council approved phase II of the project, two CAP1000 units. Construction of units 3&4 commenced in June 2022 and March 2023, respectively.

In February 2023 CNNC's Mechanized Engineering Branch subsidiary won a contract to carry out excavation and preliminary site infrastructure earthworks for units 5&6, which is anticipated to host two CAP1000s.

An agreement was signed in June 2023 for the Taizhou Nuclear Energy Zero Carbon Smart Steam Supply Project. Under the agreement the Sanmen nuclear plant would be linked to the Rongsheng Taizhou new material project industrial park via pipe allowing the transfer of 10 million tonnes of industrial steam annually.

In December 2023 the reactor pressure vessel was installed at unit 3.

In April 2025 the State Council approved phase III of the project, two Hualong One units.

Haiyang

Shandong Nuclear Power Company (a subsidiary of CPI, now SPIC) signed contracts with Westinghouse and Shaw for two AP1000 units in July 2007. China Nuclear Engineering & Construction Group (CNEC/CNECC) is building the plant. First concrete was poured in September 2009 for unit 1 and June 2010 for unit 2. The 5000 cubic metre basemat of each was placed in a single pouring of less than 48 hours. The reactor pressure vessel (RPV) and steam generators for unit 2 are being made in China. MHI supplied the turbine generators for both units. These units were expected to commence operation in December 2014 and March 2015, but in March 2014 NNSA said that start-up would be delayed into 2015 due to design changes and component problems. However, problems with the US-supplied main coolant pumps then delayed grid connection to August 2018 for unit 1, and October 2018 for unit 2. Construction time was initially expected to be 63 months, but in fact lasted 110 months for unit 1, and 100 months for unit 2. These units each deliver 1170 MWe net. See section on Embarking upon Generation III plants above.

The site will eventually have six or eight units, and in March 2009, the NDRC approved preliminary works for units 3&4, designated CAP1000 demonstration units, at the CPI site. In February 2014 the MEE (then MEP) approved construction start on these, which are expected to cost CNY 31.4 billion ($5.1 billion). The NNSA gave approval for phase 2 in December 2015. Construction start was at the time expected late 2017. Pressure testing on the RPV was completed in mid-2017. In April 2022 the State Council approved construction of Haiyang 3&4 (CAP1000 units). Construction of unit 3 began in July 2022. Construction of unit 4 began in April 2023.

The plant provided local district heating from December 2019 and it started supplying heat on a commercial basis to Fengyuan Thermal Power in November 2020. Since 2020 it has supplied the entire heating demand of Haiyang City, 20km away, and heat supply to the city of Ruchan began in 2021. In February 2023 construction began on a 23-kilometre-long heating pipe network that would transport nuclear-generated heat from the plant. The project was put into operation in 2023 and will provide an annual heating capacity of up to 9.7 million gigajoules. In November 2025 Rongcheng became the third city in the Shandong province to receive district heating from the Haiyang plant.

Haiyang will be a SPIC training base for AP1000 & CAP1000 staff, along with a set-up at Yantai.

In November 2025 SPIC began supplying heat to the city of Rongcheng in China using steam from the Haiyang nuclear power

plant.

Taishan

The first two EPRs at Taishan in Guangdong province form the main part of an €8 billion contract signed by Areva and China General Nuclear Power Group (CGN) in November 2007. The Taishan project (sometimes referred to as Yaogu) is owned by the Guangdong Taishan Nuclear Power Joint Venture Company Limited (TNPC), a joint venture between EDF (30%), CGN Power (51%) and Guangdong Yuedian Group (19%). First concrete was poured in October 2009 though officially construction – by CGN Engineering – started in December 2009. Unit 1 was commissioned in June 2018, after 104 months construction, to deliver 1660 MWe net, using a low-speed turbine. Unit 2 June 2019. In announcing the delayed start, CGN said: "The systematic integration and performance testing of nuclear power generating units of third generation nuclear technology have yet to be verified, as no such nuclear power generating unit has been put into commercial operation across the world." Unit 1 entered commercial operation in December 2018 and unit 2 in September 2019. Areva sourced the main components for both units from Japan and China. See section on Embarking upon Generation III plants above.

In April 2025 the State Council approved units 3&4 at the plant to be built as Hualong One.

Site works are reported to be proceeding for units 3&4.

Shidaowan HTR-PM

A demonstration high-temperature gas-cooled reactor (HTR) plant, with twin reactor modules driving a single 210 MWe steam turbine, was initially approved in November 2005, to be built at Shidaowan (Shidao Bay) in Weihai city, Shandong province, by Huaneng Shandong Shidaowan Nuclear Power Company Ltd (HSNPC). It was to be part of the Rongcheng Nuclear Power Industrial Park project. The HSNPC joint venture is led by the China Huaneng Group Co – the country's largest generating utility but hitherto without nuclear capacity, and still without NNSA authority to build nuclear power plants itself. However this project was classified as R&D. Huaneng Power International, with 47.5% equity, is investing CNY 5 billion in the project, which received environmental clearance in March 2008. A 20% stake in the project is held by Tsinghua

University INET, reflecting its innovative technology, and 32.5% is held by CNEC, which is promoting commercial HTR versions.

The EPC (engineering, procurement, construction) contract was let in October 2008, and involves Shanghai Electric Co and Harbin Power Equipment Co. A simulator contract signed in May 2010 was between HSNPC, Chinergy and CGNPC Simulator Co. In November 2010 Huaneng Group signed an agreement with US-based Duke Energy to train nuclear plant staff.

After three years of negotiation, in March 2011 a contract was signed with SGL Group in Germany for the supply of 500,000 machined graphite spheres to go with the HTR-PM fuel. A new HTR fuel production plant has been set up at Baotou.

With site work completed and following NDRC approval, construction of the HTR-PM started in mid-2011. NNSA approval for full construction was received early in December 2012 and first concrete followed a week later. In April 2017 the graphite moderator elements were loaded. Fuel loading began in August 2021, and first criticality was achieved in the first reactor in September, then in November for the second one. In December 2021 the unit was connected to the grid, and it began commercial operation in December 2023. In March 2024 a district heating system from the HTR-PM was commissioned, providing heating to about 1850 households and replacing some 3700 tonnes of coal per heating season.

NB: though involving twin reactors this is shown as a single reactor unit in World Nuclear Association tables.

This will be the demonstration plant for a larger commercial plant at Ruijin, Jianxi province. (See below, also Research and development section in page on China's Nuclear Fuel Cycle.)

The HTGR Industrial Alliance was launched in December 2025 by CNNC subsidiary Chinergy, bringing together more than 60 organizations to promote industrial applications of HTGR technology.

Shidaowan CAP1400 units

In November 2007, China Huaneng Group (CHNG) signed an agreement with CGN for the Huaneng Nuclear Power Development Co Ltd to build four CPR-1000 reactors at Shidao Bay, Rongcheng, Weihi city, in Shandong province in an $8 billion deal. A letter of intent regarding the first two was signed in 2008. The project then

became focused on the CAP1400 ('Guohe One'), with SNPTC as the lead partner and two CAP1400 reactors as phase 1.

The State Nuclear Power Demonstration Plant Company (SNPDP) is a joint venture company set up in December 2009 by SNPTC (55%) and China Huaneng Group (CHNG – 45%) to have overall responsibility for building and operating the first CAP1400 reactors and subsequent CAP1700 reactors (see Reactor technology section above).

The environmental impact assessment report, the site safety assessment report and construction application were submitted to the Ministry of Environment Protection and NNSA in March 2012. Site works for the first two CAP1400 units totalling 3.24 GWe were completed by April 2014. First concrete for unit 1 was initially expected in April 2015, with 56-month construction giving operation in December 2019. Unit 2 first concrete was due in August 2015 and operation in 2020. Satellite imagery shows unit 1 under construction by April 2019 and unit 2 under construction by June 2020, but construction has not been officially confirmed. Shidaowan Guohe 1 achieved commercial operation in 2024.

In April 2021 SPIC announced a 'smart energy project' for the site, linking the two Guohe One reactors with downstream applications including desalination and widespread district heating.

Shidaowan Hualong One units

In June 2014 the Ministry of Environment Protection was seeking comment on SNPTC plans for CAP1400 units 3&4, expected to cost CNY 42.3 billion. However, Huaneng, which owns the site, is now building Shidaowan 1&2 as Hualong One units, with a further two units planned (either as Hualong One or CAP1000).

The government approved the construction of Shidaowan 1&2 in July 2023. First concrete was poured for Shidaowan 1 in July 2024, and for unit 2 in May 2025.

Bailong

This is a SPIC project about 30 km southwest of CGN's Fangchenggang in Guangxi province. Bailong 1&2 will be CAP1000 units. Four CAP1400 units will be built in stage 2. In October 2020 SPIC published an environmental impact assessment for public

comment, which was approved in May 2023. Cost of stage 1 is estimated at CNY 39 billion. Construction start was originally planned for December 2021, with unit 1 commissioning in August 2026 and unit 2 in June 2027. In December 2024 SPIC announced it had begun groundworks for unit 1. In March 2025 nine major contracts were signed for the construction of units 1&2. The contracts were awarded by Guangxi Nuclear Power Company Ltd – a subsidiary of State Power Investment Corporation (SPIC) – to the Shanghai Nuclear Engineering Research and Design Institute (SNERDI) – also a subsidiary of SPIC. In September 2025 groundworks for unit 2 began. First concrete for unit 1 was poured on 25 December 2025.

Huizhou/Taipingling

This is a CGN project in Huangbu town, Huidong county, Huizhou city on Red Bay close to Daya Bay, originally with two AP1000 reactors in its first stage and four more to follow, though these will now be all Hualong One. Initial investment by CGN and Huizhou city was estimated at CNY 80 billion in 2011. The project company is China Guangdong Huizhou Nuclear Power Co, a CGN subsidiary. The project passed its initial feasibility assessment by the provincial Development and Reform Commission in 2011. Preliminary approval was reported in February 2019 and first concrete was poured in December 2019 for unit 1. First concrete for unit 2 was poured in October 2020. Hot testing was completed for unit 1 in September 2024. Fuel loading began in January 2026 and the unit was connected to the grid in February 2026. Hot testing of unit 2 was completed in August 2025.

In December 2023 construction of units 3&4 at Taipingling nuclear power plant was approved by the State Council. First concrete was poured for unit 3 in June 2025.

Lianjiang

This is a SPIC project in Guangdong which had been brought forward, with two CAP1000 units in phase 1. Construction start was initially expected in 2017. Four CAP1000s are envisaged for phase 2. Excavation works for units 1&2 began in September 2022, and construction commenced on unit 1 in September 2023 and unit 2 in April 2024. Four CAP1000s are envisaged for phase 2.

In July 2025 the reactor pressure vessel was installed at unit 1.

Tianwan Phases II, III & IV

Jiangsu Nuclear Power Corporation (JNPC) was established in 1997 to construct and operate the Tianwan nuclear plant at Tianwan in Lianyungang city of Jiangsu province. Ownership is CNNC 50%, CPI 30% and Jiangsu Guoxin Asset Management Group 20%.

Phase II, units 3&4. In October 2006, a preliminary agreement for two further 1060 MWe AES-91 VVER reactors as the second construction phase was signed with Russia's Atomstroyexport. Preliminary approval from the NDRC was received in August 2009, but protracted discussion on pricing for the Russian components of the plant delayed the project. In September 2010 a contract for the engineering design was signed with Atomstroyexport, followed by final approval from the NDRC in January 2011. In August 2013 the China Development Bank signed a loan agreement with Jiangsu Nuclear Power Corporation for CNY 16 billion for phase II.

Atomstroyexport provided 30% of the V-428M phase II units for €1.3 billion, including nuclear island equipment (reactor, steam generators, pressurizers, primary piping. etc.) and some related equipment, but did not act as the principal contractor, though it retained intellectual property rights. Jiangsu Nuclear Power was responsible for about 70% of the project, namely, the civil work, turbine island with equipment and related infrastructure onsite. The EPC contract with CNNC's China Nuclear Engineering & Construction Group (CNEC/CNECC) was signed in October 2011. A civil engineering contract was let to China Nuclear Industry Huaxing Construction Company (HXCC) in May 2012, and to China Nuclear Industry 23 Construction Co for component installation in July. Both are CNEC subsidiaries. Iskorskiye Zavody, part of Russia's OMZ, supplied the major components including two reactor pressure vessels with internals and upper units. (The company already took part in making the major equipment for Tianwan 1&2, including reactor pressure vessels.) ZiO-Podolsk made the steam generators. The turbine generator sets are from Dongfang Electric, using Alstom Arabelle low-speed technology. Areva I&C systems are being used.

First concrete for unit 3 was poured in December 2012, and that for unit 4 in September 2013. Unit 3 started up in September 2017, was grid connected at the end of December, and entered commercial operation in February 2018. Unit 4 achieved first criticality in September 2018 and grid connection in October after

61 months' construction. It entered commercial operation in December 2018.

In May 2022 CNNC commenced work on the 'Heqi 1' project at units 3&4 of the Tianwan plant to supply 4.8 million tonnes of industrial steam annually to the Lianyungang Petrochemical Industrial Base. It was commissioned in June 2024.

There was some uncertainty about the technology for **phase III**, **units 5&6.** Due to urgency in meeting power demands, it appeared likely that units 5&6 (phase III) might be built ahead of 3&4, as CPR-1000 units, based on those at Fangjiashan. This technology prospect appeared to lapse with post-Fukushima sidelining of the design. But when finally approved by the State Council in December 2015 it was confirmed that they would be the last of the M-310 based units, CPR1000. An EPC contract between Jiangsu and CNPE was signed in February 2011, making CNPE the project manager. CNPE signed a contract for both nuclear islands in June 2016 with China Nuclear Industry Huaxing Construction Co (HXCC). Dongfang Electric has a contract to supply turbine generators using Alstom Arabelle low-speed technology. Construction of unit 5 started in December 2015, and of unit 6 in September 2016. The NNSA issued an operating licence for unit 5 in July 2020, and two days later CNNC announced the completion of loading of the first core. First criticality at unit 5 was achieved on 26 July 2020 and it was connected to the grid on 8 August. Cold testing of unit 6 was completed in October 2020; it was connected to the grid in May 2021, and entered commercial operation in June 2021.

Phase IV, units 7&8, was less ambiguous. In December 2012 there were discussions with CNNC regarding phase IV of Tianwan. In October 2013 China National Nuclear Power Co Ltd (CNNP) said that the technology might be VVER-1200/AES-2006 rather than VVER-1000/AES-91. In November 2016 a new cooperation agreement with Russia was announced including Tianwan 7&8. In June 2018 Rosatom announced that contracts had been signed for Russia to supply two VVER-1200 units at Tianwan as units 7&8. The contract also includes a deal for two further VVER-1200 units at Xudabao (units 3&4), that were originally planned to be of CAP1000 design. General contracts for construction of Tianwan 7&8 were signed in March 2019, and for Xudabao 3&4 in June 2019. In July 2019 TVEL announced it had signed a contract to supply fuel for Tianwan 7&8. Construction commenced at Tianwan 7 in May 2021, followed by unit 8 in February 2022. China Nuclear Industry No.23 Construction Co Ltd is contracted to build the nuclear islands of units 7&8. The V-491 reactors are based on those at the Leningrad power plant.

In October 2023 the reactor pressure vessel for unit 7 was installed. Cold testing of unit 7 was completed in July 2025, followed by hot testing in January 2026.

Changjiang

CNNC's Changjiang nuclear power plant on Hainan Island started construction in April 2010 for operation in 2015. Total cost of the first pair of CNP-600 units based on those at Qinshan phase II is put at about CNY 20 billion ($3.15 billion). Unit 1 criticality was in October 2015, grid connection in November and commercial operation in December. Unit 2 was grid connected in June 2016 and started commercial operation in August.

Construction of units 3&4, both Hualong One, began in March and December 2021 respectively. Huaneng Power International (HPI), part of China Huaneng Group (CHNG), holds a 49% share in Hainan Nuclear Power Co Ltd. with CNNC 51% for phase 1 (units 1&2). For phase 2 (units 3&4) the Huaneng Hainan Changjiang Co is the investment vehicle which is 51% owned by Huaneng Nuclear Power development company and 49% by CNNC. China Huaneng Nuclear Development Corp. will manage construction. More than 80% of the plant's equipment is made in China. The inaugural ceremony for these was in November 2019. Construction time of each unit is expected to be 60 months, and the cost of both is given as CNY 39.45 billion. China Nuclear Industry No.5 Construction Company is building the Changjiang 3&4 nuclear islands. The reactor pressure vessel for unit 3 was installed in September 2023.

Two demonstration multi-purpose modular ACP100 'Linglong One' units will now be built at Changjiang (instead of Putian in Fujian province). This is China Guodian's first nuclear power venture, with CNNC holding 51% of CNNC New Energy Corporation (CNNC-CNEC). The ACP100 units are integral PWRs, 125 MWe, with passive cooling for decay heat removal. CNNC said that the units could provide electricity, heat and desalination. Construction time is expected to be 58 months. It involves a joint venture of three companies for the demonstration plant: CNNC as owner and operator, the Nuclear Power Institute of China as the reactor designer and China Nuclear Engineering Group being responsible for plant construction. Unit 1 commenced construction in July 2021. Cold functional tests were completed in October 2025, followed by the non-nuclear turbine test run in December 2025. Commercial operation is expected in 2026, which would make it the world's first commercial land-based SMR.

In 2019 Hainan Province Development and Reform Commission published energy policies specifying that nuclear power will become the primary source of electricity for the island, with gas power mainly used for peaking management, and no more coal power plants to be built.

Xudabao

CNNC's Xudabao (or Xudapu) nuclear power station is in Xingcheng City, Huludao (Hulu island), in coastal Liaoning province. CNNC's Liaoning Nuclear Power Company Ltd owns the plant, 70% CNNC with Datang International Power Generation Co holding 20% equity, and State Development and Investment Corporation (SDIC) 10%. The general contractor is China Nuclear Power Engineering Company Ltd (CNPE), a subsidiary of CNNC. In October 2010, the Northeast Electric Power Design Institute (NEPDI), Changchun, Jilin, a subsidiary of China Power Engineering Consulting Corporation (CPECC), signed a survey and engineering contract for the plant.

The CNY 110 billion ($18 billion) Xudabao project was originally expected to comprise six CAP1000 reactors, with units 1&2 in the first phase. Site preparation was under way in November 2010, and NNSA granted site approval for two units in April 2014. Final approval was reported in September 2014, and again in December 2015 by the NNSA. However, in June 2018, it was announced that two VVER-1200/V-491 units would be delivered by Russia at Xudabao (3&4). In September 2018, Rosatom said it hoped to commission the two units in 2027-2028. Rosatom would supply the nuclear islands, with turbine generators and balance of plant being Chinese. A $1.7 billion contract between Atomstroyexport and CNNP and CNEIC was signed, with a view to 69 months construction. In November 2019 TVEL announced it had signed a contract to supply fuel for Xudabao 3&4 and in February 2021 AEM reported it had started manufacturing the main nuclear equipment for unit 3. Construction commenced on unit 3 in July 2021, and on unit 4 in May 2022.

In October 2016 an EPC contract for construction of Xudabao 1&2 nuclear islands was signed with China Nuclear Industry 22 Construction Co, a CNECC subsidiary. Units 1&2 will be CAP1000 units with a high rate of localization.

When Taohuajiang plant was deferred, the main AP1000 reactor parts were transferred to Xudabao.

The State Council approved construction of units 1&2 in July 2023, and first concrete was poured for unit 1 in November 2023.

Lufeng

CGN Lufeng Nuclear Power Corporation is constructing the first of four CAP1000 reactors at Lufeng (Shanwei) plant in the Tianwei district in eastern Guangdong. An EPC contract was signed with SNERDI in September 2013, environmental approval was in June 2014, and NDRC approval for two units was reported in September 2014. The NNSA gave approval in December 2015. Commercial operation was expected in 2020-21, but the state council did not approve the project until August 2024.

In September 2025 SNERDI announced the installation of the 'CA01 super module'. The CA01 module sits inside the unit's containment module where it will house the plant's reactor pressure vessel, steam generators and other components. The CAP1000 reactor design – the Chinese version of Westinghouse's AP1000 – uses modular construction techniques, enabling large structural modules to be built at factories and then installed at the site.

Construction of unit 1 began in Feburary 2025, followed by unit 2 in December 2025.

CGN is also constructing two Hualong One reactors (as units 5&6) at the site, approved by the State Council in April 2022. First concrete was poured for unit 5 in September 2022 and for unit 6 in August 2023.

Putian & Zhangzhou-Gulei

China Guodian's first nuclear power venture, with CNNC holding 51% of CNNC New Energy Corporation, was to involve two small modular ACP100 reactors, at Xinghua Bay, Putian, Zhangzhou city, on the coast in Fujian province, near Xiamen, as a demonstration plant. These are integral PWRs, with passive cooling for decay heat removal. This was to be the CNY 5 billion ($788 million) phase 1 of a larger Zhangzhou project (see below). However, the ACP100 plan has now been moved to Changjiang.

A second proposal is for two ACP100 units on the other side of the peninsula, at Zhangzhou-Gulei, but though approved by SASAC in November 2011, it is stalled due to opposition from local

government. CNNC New Energy Corporation (CNNC & Guodian) is seeking alternative sites.

Zhangzhou

The larger project in Fujian is being undertaken by CNNC subsidiary China National Nuclear Power Co (CNNP) and will comprise two Hualong One reactors as phase 1 and two more as phase 2. In May 2014 the local government gave approval, with phase 1 costing CNY 54.4 billion, though in 2020 with both under construction the cost was quoted at CNY 45 billion. The NNSA gave approval in December 2015, for AP1000 units and confirmed site selection in October 2016. However, CNNC decided to use the Hualong One design here. At the start of October 2019 the Ministry of Ecology & Environment issued a construction licence, and on 16 October construction commenced on the first unit.

The plant owner, CNNC Guodian Zhangzhou Energy Co. Ltd was established in November 2011, by CNNC or China Nuclear Power International Inc (51%) and China Guodian Corporation (49%). The proposal was submitted to the NDRC in August 2010. The NEA announced preparatory work on units 1&2 for 2017. Late in 2016 the KSB Group in Germany was awarded a contract for six reactor coolant pumps for units 1&2, to be delivered in 2020 and 2021. In mid-2017 China Nuclear Industry No.24 Construction Company won the contract for the nuclear island civil engineering. In February 2019 CNNC subsidiary China National Nuclear Power (CNNP) released its environmental impact assessment for public comment. First concrete was poured for unit 1 in October 2019, and for unit 2 in September 2020.

Zhangzhou 1 was connected to the grid in November 2024 and began commercial operation in January 2025. In October 2025 fuel loading began at Zhangzhou 2. Unit 2 was connected to the grid in November 2025 and began commercial operation in January 2026.

CNNC issued the environmental impact assessment for units 3&4 in October 2020. First concrete was poured for unit 3 in February 2024, and for unit 4 in September 2024.

In November 2022 CNNC began construction of China's first nuclear generation-supported pumped hydro storage project, powered by the six planned reactors at Zhangzhou. The project is expected to be completed in just under six years.

Hongshiding (Rushan)

In November 2006, an agreement was signed by CNNC to proceed with the first two units of the Hongshiding nuclear plant at Hongshiding in Weihai or Rushan city, Shandong province, costing $3.2 billion, with construction to begin in 2009 and first power in 2015. However, it has been deferred or possibly cancelled due to public acceptance issues. Six units now likely to be Hualong One, 1150 MWe each, are envisaged at the site, with Shandong Hongshiding Nuclear Power Co. Ltd as developer.

Xuwei, Lianyungang

CGN has several plans in Northern China, and among other possible projects it had applied for several reactors at Lianyungang, Jiangsu province, very close to CNNC's Tianwan plant. A proposal was submitted to the NRDC.

However, in August 2024 the State Council approved phase 1 of CNNC's Xuwei plant in Lianyungang. Suneng Nuclear Power Company, a CNNC subsidiary, is the owner of the Xuwei project and responsible for project investment, construction and operation management.

The plant will supply both industrial heating and electricity by coupling in its first phase two Hualong One units and one high-temperature gas-cooled reactor (HTR-PM600). Steam from the Hualong One units will be used to heat demineralized water to produce saturated steam, and the primary steam of the HTGR will be used to heat the saturated steam for the second time. The project is expected to supply 32.5 million tonnes of industrial steam annually.

Construction of unit 1 commenced in January 2026.

Songjiang

This is to be developed in Shanghai's Songjiang district by CGN with China GD Power Development Co Ltd, a subsidiary of Guodian, and in connection with a framework cooperation agreement with the State Grid Corporation of China.

Xiangshan Jinqimen

Construction of two Hualong One units at a site near Hepu Town in Zhejiang province was approved by the State Council in December 2023. Pre-feasibility environmental and seismic survey work began at the site in 2014. In August 2025 first concrete was poured for unit 1. The site is about 50 nautical miles from the Sanmen nuclear power plant. The site was originally planned to host AP1000 units. A further four Hualong One units are proposed beyond the first two.

Haixing

This will be the first nuclear plant in Hebei province, 200 km from Beijing, Hebei being the last coastal province to host a nuclear plant. CNNC CHD Hebei Nuclear Power Corporation (CCHNPC) is owned 51% by CNNP, 39% by Huadian Power International Corporation and 10% by Hebei Construction & Investment Group and will be responsible for the construction, operation and management of the plant. Construction was originally planned for June 2016 for unit 1 and ten months later for unit 2. Planned construction time is 54 months. The site has scope for 7500 MWe of plant. The project investment is estimated at CNY 120 billion, with CNY 45 billion for this first phase. In May 2023 preparatory work was under way.

Sanming

In October 2009, an agreement was signed by CIAE and CNEIC (a CNNC subsidiary responsible for technology imports) with Russia's Atomstroyexport to start pre-project and design works for a commercial nuclear power plant with two BN-800 fast neutron reactors (referred to as Chinese Demonstration Fast Reactors) at Sanming city, an inland part of Fujian province. A site survey and preliminary feasibility study had been undertaken in 2007-08. CNNC in April 2010 established Sanming Nuclear Power Co Ltd as a joint venture company with the Fujian Investment & Development Corp and local government, and initiated a full feasibility study.

Construction was due to start in 2013, once an intergovernmental agreement was in place, expected in 2012. The local content was targeted at 70%, and the first unit was to be in operation in 2018, with the second following about a year later. A second phase, with units 3&4, was due to commence in 2015. The plant would be

similar to the OKBM Afrikantov BN-800 design built in Russia at Beloyarsk 4, which started up in 2014, though this is actually an experimental reactor rather than commercial type. Negotiations on price earlier delayed the project and NIAEP-Atomstroyexport then said it expected the first contracts to be in place at the end of 2014. However, there is no indication of BN-800 plans proceeding. The project is superseded by the CFR600 fast reactor.

Xiapu

In 2015 China National Nuclear Power Company Limited Xiapu was set up in Fujian, with 55% CNNC, 20% Fujian Funeng Co Ltd, 10% Huaneng Nuclear Power Development Corp, 10% China Yangtze Power Co Ltd and 5% Ningde State-owned Assets Investment Management Co. The plan was to build the first 600 MWe Terrapower so-called travelling wave fast reactor there, the TWR-P, but two demonstration CFR600 reactors are now being built at the site, by China Nuclear Industry 23 Construction Co.

China Huaneng Group (CHG) is to build four Hualong One units at the site. In August 2022 the company said it expected to receive full government approval in either 2022 or 2023, with construction start a year later. In December 2024 the environmental impact assessment for the project was approved. CHG also plans to build an HTR600 unit at the site.

Cangnan/San'ao

CGN has been progressing plans for a six-unit plant near Cangnan county on the coast of Zhejiang province since 2017. Initially, two AP1000s were planned for the site, but the units will now be Hualong One.

CGN holds a 46% stake in the Cangnan Nuclear Power project company, with Zhejiang Electric Power Co (34%), Cangnan Haixi Construction Development Co (9%), Wenzhou Nuclear Energy Development Co (9%), and Geely Maijie Investment Group (2%). The Geely share is the first time that local private capital has been involved in a Chinese nuclear plant*. The plant has been referred to as both Cangnan and San'ao or Sanao nuclear power plant.

* Hong Kong utility China Light & Power (CLP) has equity in CGN's Daya Bay and Yangjiang plants.

In April 2020 the NNSA and the Ministry of Ecology and Environment (MEE) published the site selection and environmental impact assessment reports for the two units for public comment. In July 2020 the MEE accepted the environmental impact statement. Total investment for the first two units is estimated at CNY 39.3 billion ($5.5 billion). Construction of unit 1 started on the last day of 2020 and that of unit 2 on the last day of 2021.

In February 2023 the MEE announced that it plans to approve the environmental impact assessment documents covering the site selection stage for San'ao units 3&4. It was then expected that unit 3 would start construction at the end of 2023/start of 2024, with construction beginning at unit 4 ten months after.

In March 2023 Zhejiang Zheneng Electric Power Co. and CGN announced a joint venture for the development of units 3&4, referred to as "CGNPC Cangnan Second Nuclear Power Co., Ltd.". The shareholders are: CGN (51%), Zheneng Electric (31%) Wenzhou Nuclear Energy (8%), Huanan County (8%), and Zhejiang Geely Industry Investment (2%).

Cold testing of unit 1 was completed in November 2024, followed by hot testing in June 2025 and fuel loading in December 2025. Unit 1 was connected to the grid in March 2026. Cold testing of unit 2 was completed in October 2025.

In November 2025 CGN poured first concrete for unit 3.

Zhaoyuan

In November 2022 the Ministry of Ecology and Environment (MEE) announced the formal acceptance of the environmental impact assessment documents covering the site selection stage for Zhaoyuan nuclear power plant (six Hualong One units) located in Shandong province, eastern China. Although the site is located about nine kilometres from Laizhou Bay, it will be seawater-cooled, so it is considered to be a coastal site. Units 1&2 were approved by the State Council in August 2024.

In November 2025 CGN poured first concrete for unit 1.

Inland nuclear power plants

These have been delayed significantly from the dates for the larger units planned before mid-2011. This was announced as being due

to concerns regarding possible pollution of rivers. Taohuajiang, Xianning and Pengze are all in the 12th five-year plan to start construction, but the premiere's announcement in October 2012 deferred approvals for inland plants until after 2015. The delay has involved an assertion of central government control over eager provincial governments with their own economic development agendas. NEA approvals were initially expected to start for these in 2016.

The State Council in 2008 prioritized three projects, and subsequently these remain as likely first construction starts for 13th Five-Year Plan from 2016: Taohuajiang in Hunan, Pengze in Jiangxi, and Dafan in Hubei. In a formal revision of the Five-Year Plan in mid-2018, the Ningxia Hui Autonomous Region was added with possible construction start for 2 GWe by 2020. However, no approval for inland plant construction has been forthcoming. The 15th Five-Year Plan (2026-2030) continues to restrict nuclear development to coastal sites.

In June 2015 CNNC signed a strategic cooperation agreement with Sichuan provincial government to promote “large-scale commercial nuclear power reactors and renewable energy generation, nuclear power equipment manufacturing, nuclear technology and innovation platform construction.” Sichuan had earlier promoted Nanchun as a power plant site, and CGN is involved with this – see below.

In September 2015 it was reported that 31 inland sites had cleared preliminary approval by NDRC, and awaited State Council approval.

Taohuajiang

CNNC's Taohuajiang nuclear power plant, also referred to as the Taohua [peach blossom] River project, on the Zi River in Taojiang county,Yiyang city, near Yueyang in inland Hunan province was planned to be China's first inland nuclear power plant. It was expected to start construction in September 2010, and some CNY 3 billion of site works have been carried out, but construction was cancelled. CNNC set up Hunan Taohuajiang Nuclear Power Co Ltd. to build and operate the plant – a four-unit CAP1000 project costing over CNY 70 billion.

The project was approved by the NDRC in November 2005, and in 2008 the project was approved for preliminary construction. Site works to the extent of CNY 3.8 billion were undertaken. The design

by SNERDI under SNPTC and SNPDRI was submitted to the NNSA in February 2010 for licensing. A general framework agreement for construction was signed by CNNC with CNPE Corporation as EPC contractor in December 2010. The first unit was originally expected in commercial operation in 2015, and the fourth in 2018. However, after all the inland projects were deferred the reactor components were transferred to Xudapu/Xudabao, and the site workforce was laid off.

Xianning/Dafan

In August 2008, CGN and Hubei Energy Group Ltd set up the Hubei Nuclear Power Company as a joint venture and announced plans to build a nuclear power plant at Dafan in Xianning city of the inland Hubei province. Site works for this plant (four AP1000 units) have been undertaken to the extent of CNY 3.4 billion. Construction of the first two units was expected to start in 2011. The reactor pressure vessel for the first unit is contracted to China First Heavy Industries, and the first two 209 metre high cooling towers to Belgium's Hamon Thermal. The cost of four AP1000 reactors was put at CNY 60 billion ($8.8 billion). This would have been CGN's first AP1000 plant, but the equipment has been reassigned to Lufeng and now CAP1000 units are planned. A further phase is estimated to cost CNY 45 billion.

The large pre-assembled modules that will make up the bulk of the new CAP1000s are to come from a new inland facility owned by new firm Hubei Nuclear Power Equipment Company.

Reports of a Songzi plant may refer to later stages of Dafan, though possible projects in Yangxin county have been mentioned.

The Hubei Nuclear Power Co is also reported to be planning a four-unit CAP1000 plant at Guangshui city in the northeast of the province.

Pengze

CPI's Pengze Nuclear Power Project in Jiangxi province was to have four AP1000 reactors costing CNY 60 billion ($8.8 billion). The site has been prepared for the first two units, and safety and environmental approvals were obtained in May 2009. CPI signed the EPC contract framework for phase 1 (units 1&2) in August 2009, the engineering project contract was reported to be between CPI Jiangxi Nuclear Power Co Ltd and CPIC. The equipment

procurement was reported to be between CPIC and China Power Complete Equipment. CPI aimed to start construction in 2010, for 2013 start-up, but construction has been delayed, evidently to 2018, and CAP1000 units are now planned. Site works amount to CNY 3.4 billion.

The project is inland in Juijiang city, across the Yangtze River from Wangjiang in Anhui province. The cooling towers are being designed by Belgium's Hamon Thermal for the State Nuclear Electric Power Planning Design and Research Institute (SNPDRI). The project has been opposed by Wangjiang in neighbouring Anhui province, which has plans for several nuclear plants iteself, including Wuhu (Fanchang) and Jiyang (Chizhou), with Anqing Congyang and Xuancheng along the Yangtze River also mentioned.

Wuhu

The Wuhu nuclear plant on the Yangtze River in the Bamaoshan area, Fanchang county, of Anhui province was planned to have four 1000 MWe CPR-1000 units, but is now designated for CAP1000s to be constructed in two phases. CGN's proposal for two units of phase 1 has been submitted, some preparatory work has been undertaken and the Anhui Wuhu Nuclear Power Co has been set up, with 51% CGN ownership. The environmental impact statement was released for public comment in January 2010.

Ningdu/Ganzhou

CNNC New Energy Corp, the joint venture of CNNC (51%) and Guodian Corp, has also signed a preliminary agreement for small modular reactors with Ganzhou city in Jiangxi province.

Xiaomoshan

The Xiaomoshan nuclear power plant on the Yangtze River in Huarong county, Yueyang city, Hunan province (inland), is a priority project for CPI. It will eventually have six CAP1000 reactors and be built by Hunan Nuclear Power Company Ltd in two phases. NDRC approval was given in 2006 but as of mid-2010 NNSA approval was awaited. Site preparation has been undertaken and first concrete was expected late in 2010 but has not proceeded. The cost is put at CNY 70 billion ($10.25 billion) for the first four units, funded by SNPTC and Wuling Electric Power Development Co. (a

CPI subsidiary). The Heimifeng pumped storage plant will be associated with it.

Yanjiashan/Wanan/Ji'an

In August 2009, CNNC (51%) signed a joint venture agreement with Jiangxi Ganneng Co. Ltd and Jiangxi Ganyue Expressway Co Ltd (49% between them) setting up Jiangxi Nuclear Power Co to build the Wanan Yanjiashan nuclear power project at Ji'an in the Jiangxi province. CNNC contracted a feasibility study of Yanjiashan nuclear power program in July 2010. Pre-project work was reported as under way in November 2010. (This is also reported as a CPI project.)

Shaoguan

CGN's Shaoguan nuclear plant will comprise four CAP1000 reactors and is expected to cost RMB 50 billion. It will be located in Baitu Town of Qujiang District in Shaoguan City, and will be the first inland nuclear power project in Guangdong. The Shaoguan Nuclear Power Co was established in April 2010.

Zhongxiang

CNNC's Hubei Zhongxiang nuclear power project is at Zhongxiang city in central Hubei, with China Datang. The 5000 MWe plant is undergoing a detailed feasibility study, but further details are unknown.

Hengfeng/Shangrao

CNNC New Energy Corp, the joint venture of CNNC (51%) and China Guodian Corp, has also signed an agreement for small modular reactors with Hengfeng county, Shangrao city in Jiangxi province.

Nanchun/Nanchong/Sanba

In 2005, Sichuan province proposed Nanchun/Nanchong city east of Chengdu as a suitable site for a nuclear power plant and sought approval for it from the National Development and Reform

Commission (NDRC), which was not given, possibly because of seismic concerns. In March 2009, the provincial government signed an agreement with CGN to pursue the plan for a Nanchun nuclear power plant, involving the Nuclear Power Institute of China (NPIC), headquartered in Chengdu. Preliminary plans in 2008 were for a 4000-6000 MWe Sanba nuclear power plant on the Jialing River, at a cost of CNY 25 billion ($3.7 billion), but four Hulaong One units are now proposed. Majority ownership would be CGN.

Another Sichuan agreement for a nuclear power plant project has been signed between CNNC and Yibin city, south of Chengdu.

Xiangtan

In December 2009, China Huadian Corp signed an agreement with Xiangtan city government in Hunan to undertake studies for a CNY 60 billion power plant comprising four 1250 MWe reactors. A refined proposal was expected in September 2010. This will apparently be the fourth nuclear project for China Huadian.

Longyou/Zhexi

In October 2008 a project proposal was submitted to NDRC by CNNC and Zhejiang Energy Group Co Ltd for a nuclear power plant in Hangzhou, western Zhejiang province, with four AP1000 reactors, though earlier reports had four 1000 MWe units to be built in two phases from late 2010. The proposed site is Tuanshi, Longyou county. Pre-project work was reported as under way in November 2010.

Jingyu

CPI plans to spend CNY 85 billion to build the six-unit Jingyu or Chisong nuclear power plant in Jingyu county near Baishan, in southern Jilin province, with four CAP1000 units to be in stage 1. The project is still in the preliminary feasibility stage, though site preparation is under way. Construction start was scheduled for 2012.

Nanyang

To be a six-unit CNNC or CPI plant in Henan province. Pre-project work was reported as under way in November 2010.

Tongren

In July 2014 Guizhou province announced that it had contracted with CGN to build two 1250 MWe reactors at Tongren city for CNY 35 billion, by 2020. The site is being decided among Dejiang, Sinan and Yanhe counties. CGN has also agreed to build two 100 MWe nuclear units for CNY 3 billion, and a 1000 MWe wind farm for CNY 5 billion.

The following inland plants have been proposed but are not understood to be under active consideration in 2015:

Jiutai, Liangjiashan

These two nuclear power plants planned for northern Jilin province close to Changchun, were to be developed by CGN with China GD Power Development Co Ltd, a subsidiary of Guodian, and in connection with a framework cooperation agreement with the State Grid Corporation of China. They are not expected to be approved before about 2020.

Jiyang

Besides Wuhu, CNNC was reported as starting a feasibility study on another four-unit nuclear plant in the Anhui province, at Jiyang in Chizhou city, in December 2008.

Changde/Hengyang

In August 2009, CNNC signed an agreement with Hengyang city in Hunan province to build a nuclear power plant there or nearby. This is about 200km south of its Taohuajiang project at Yiyang city in Hunan. China Guodian Corporation, one of the country's largest power producers, is involved in the project.

Ruijin

In April 2015 CNEC announced that its proposal for two commercial 600 MWe HTR units at Ruijin city in Jiangxi province had passed an initial feasibility review. The design is based on work by Tsinghua university, taking shape as the Shidaowan HTR-PM of 210 MWe. Each 600 MWe unit would have six small reactors driving one steam turbine. Ruijin Nuclear Engineering HTR NPP Co, Ltd was established in November 2014. Construction start was envisaged for 2017 but the project did not proceed.

Zhuanghe

The Liaoning Zuanghe nuclear power project at Nanjian, Zhuanghe city, is headed by Datang Power, with 46% equity. In December 2024 the project's environmental impact assessment was approved.

Nuclear technology exports

China has ambitions for increasing its nuclear technology exports in line with its wide geopolitical role and its industrial capabilities. However, it is limited to some extent by:

- Russia's established field of influence.
- Not yet being party to the IAEA Vienna Convention on Civil Liability for Nuclear Damage (Russia is a party).
- Not yet set up to take back the used fuel (as is normal for Russian reactor exports, and a request expressed by new nuclear countries).

China has a determined policy at NDRC level of exporting nuclear technology, based on development of the CAP1400 reactor, and subsequently the Hualong One reactor, with Chinese intellectual property rights and backed by full fuel cycle capability. The 'go global' policy is being pursued at a high level politically, utilising China's economic and diplomatic influence. In January 2015 the cabinet announced new incentives and financing for industry exports, particularly nuclear power and railways, on the back of $103 billion outbound trade and investment in 2014. In May 2017 the Belt & Road Initiative (BRI) was formally launched, with much fanfare, to boost global connectivity and trade and inviting countries to become partners in the BRI. Projects in Pakistan,

Indonesia, eastern Europe and northern Africa were mentioned along with Chinese funding of $75 billion from banks, $20 billion through a new Silk Road Fund, and $12 billion as aid.

CNNC and SPIC/SNPTC are focused on the export potential of the CAP1400, and SNPTC aims at "exploration of the global market" from 2013, particularly in South America and Asia.

The Hualong One reactor is also intended for export, with CGN focusing on Europe and CNNC elsewhere, particularly Pakistan and South America. CNNC is keen to export the Hualong One reactor more widely, and says it is open to EPC (engineering, procurement, construction), BOT (build, operate, transfer) and BOO (build, own, operate) project models. It sees potential for selling up to 30 Hualong One reactors abroad through the BRI.

Export sales and prospects for Chinese nuclear power plants

Country	Plant	Type	Est. cost	Company	Status, financing
Pakistan	Chasma 3&4	CNP-300	$2.37 billion	CNNC	Unit 3&4 operating, Chinese finance 82% of $1.9 billion, Exim-Bank. (Units 1&2, also CNP-300, earlier built by China)
	Karachi Coastal 1&2	Hualong One	$9.6 billion	CNNC	Units 1&2 operating, $6.5 billion vendor finance, maybe 82% China finance, Exim-Bank
Romania	Cernavoda 3&4	Candu 6	€7.7 billion	CGN	CGN agreement terminated January 2020. EPC contract signed November 2024 with Western consortium (Fluor, AtkinsRéalis, Ansaldo Nucleare, Sargent & Lundy) for €3.2 billion.
Argentina	Atucha 3	Candu 6	$5.8 billion	CNNC	Proposed, with 85% China finance
	5th Argentine reactor	Hualong One	$7 billion	CNNC	Proposed with 85% China finance
UK	Bradwell	Hualong One		CGN	Cancelled
Iran	Makran coast	2 x 100 MWe		CNNC	Agreement July 2015

Country	Plant	Type	Est. cost	Company	Status, financing
Turkey	Igneada	AP1000 and CAP1400		SNPTC	Exclusive negotiations involving Westinghouse, 2014 agreement
South Africa	Thyspunt	CAP1400		SNPTC	
Kenya		Hualong One		CGN	MoU July 2015
Egypt		Hualong One		CNNC	MoU May 2015
Sudan		ACP600?		CNNC	Framework agreement May 2016
Armenia	Metsamor	1 reactor		CNNC	
(No country)		HTR600		CNEC	Export intention
Kazakhstan		Fuel plant JV		CGN	Agreement Dec 2015

In Pakistan, the fourth new 300 MWe CNP-300 PWR unit has been completed at Chasma, by China Zhongyuan Engineering Corporation, joining the three built there earlier, supplied by CNNC.

In 2013 CNNC announced an export agreement for twin ACP1000 units, for Pakistan's Karachi Coastal Power station, costing $9.6 billion. This is using Hualong One technology and being built by China Zhongyuan Engineering Corporation. Construction of the first unit started in August 2015 and the second in May 2016. In July 2023 Pakistan formally approved a fifth Chashma unit, a Hualong One, with first concrete poured in December 2024.

In May 2014 Romania's Nuclearelectrica signed an agreement with CGN to explore the prospect of building two new reactors at Cernavoda, which currently has two Candu 6 reactors. In November 2013 two nuclear cooperation agreements were signed by Nuclearlectrica with CGN, one a letter of intent relating to construction of units 3&4. In November 2015 the two companies signed a further agreement for the development, construction, and operation of the two Candu 6 units. Construction was expected from about 2018 but will now be 2021 or later.

In July 2014 a high-level agreement was signed by Argentine and Chinese presidents towards construction of Atucha 3 as a PHWR

unit. CNNC will provide most of the equipment and technical services under long-term financing. Candu Energy will be a subcontractor to CNNC. In September the utility NASA signed a commercial framework contract with CNNC to progress this, with CNNC's Qinshan Phase III units as reference design for a Candu 6 unit. It will have $3.8 billion in local input and $2 billion from China and elsewhere under a long-term financing arrangement. However, under President Milei from December 2023, the project has been effectively frozen as Argentina focused on domestically-designed small modular reactors.

SNPTC is keen to export the CAP1400 reactor, and considers Turkey and South Africa to be good prospects. In November 2014 SNPTC signed an agreement with Turkey's utility EUAS and Westinghouse to begin exclusive negotiations to develop and construct a four-unit nuclear power plant in Turkey. In December 2014 it signed two agreements in South Africa with a view to nuclear power plant construction, and CNNC signed another there.

CNEC has promoted the HTR technology to Dubai, UAE, Saudi Arabia, South Africa and Indonesia, and has signed agreements with some of them.

In June 2019, the China's People's Political Consultative Conference (CPPCC) suggested that as many as 30 Chinese reactors could be built overseas by 2030 as part of the BRI.

The development of small reactors described above, led by the ACP100, is primarily with a view to export markets, following demonstration units at home.

Development of reprocessing capacity is seen to be important as a service industry backing up nuclear technology exports.

As well as specifically nuclear exports, the State Grid Corporation of China (SGCC) is building a 2084 km UHV DC transmission line in Brazil, for over $2.3 billion.

Notes & references

Notes

b. The CNP series of reactors is also referred to as the CP series. [Back]

c. The ACP600 design appears to be an advanced version of the CNP-600. [Back]

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Appendices

Appendix 1: Nuclear Organisations in China

Related information

China: Nuclear Fuel Cycle